

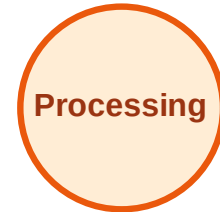
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



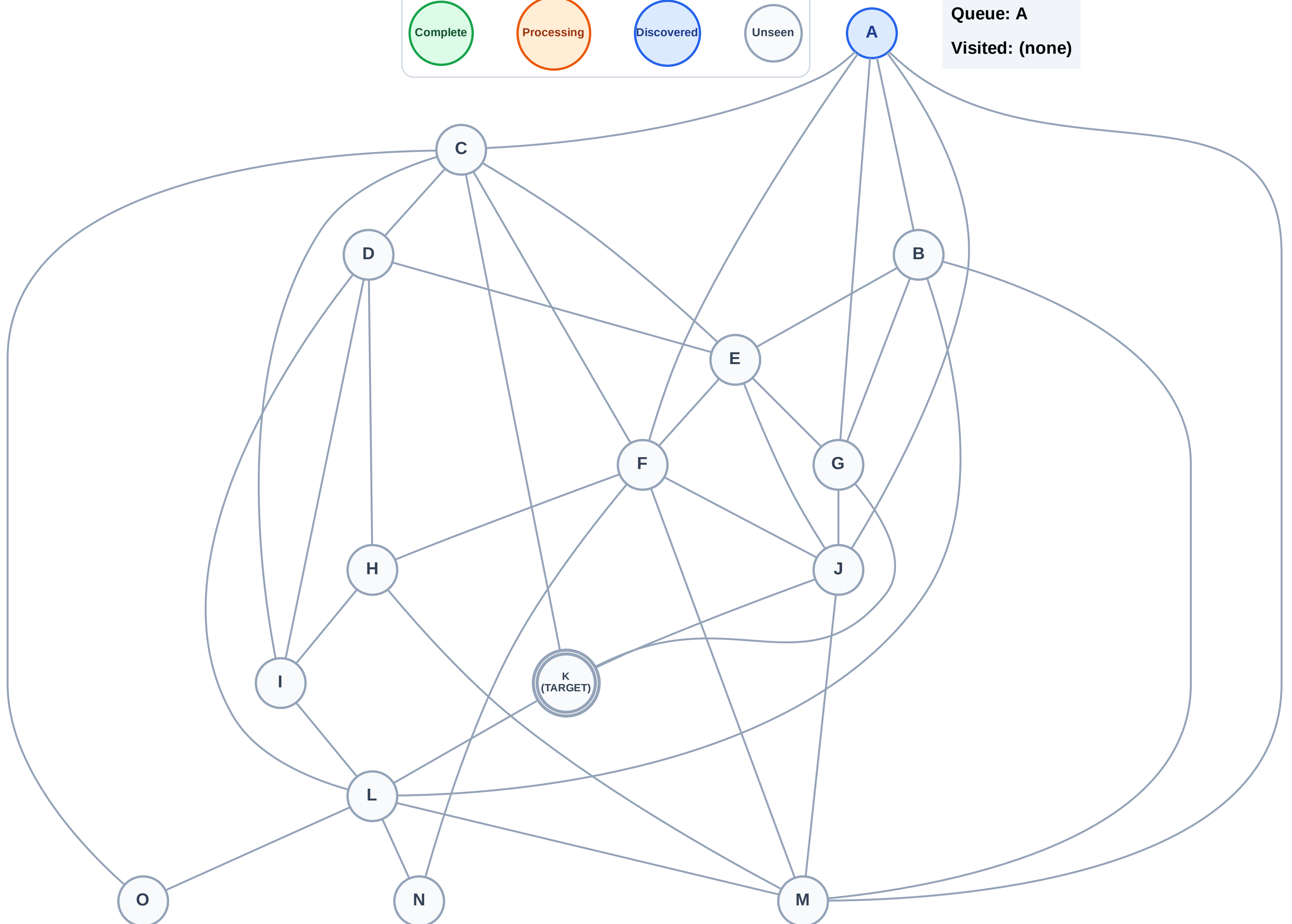
Discovered



Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



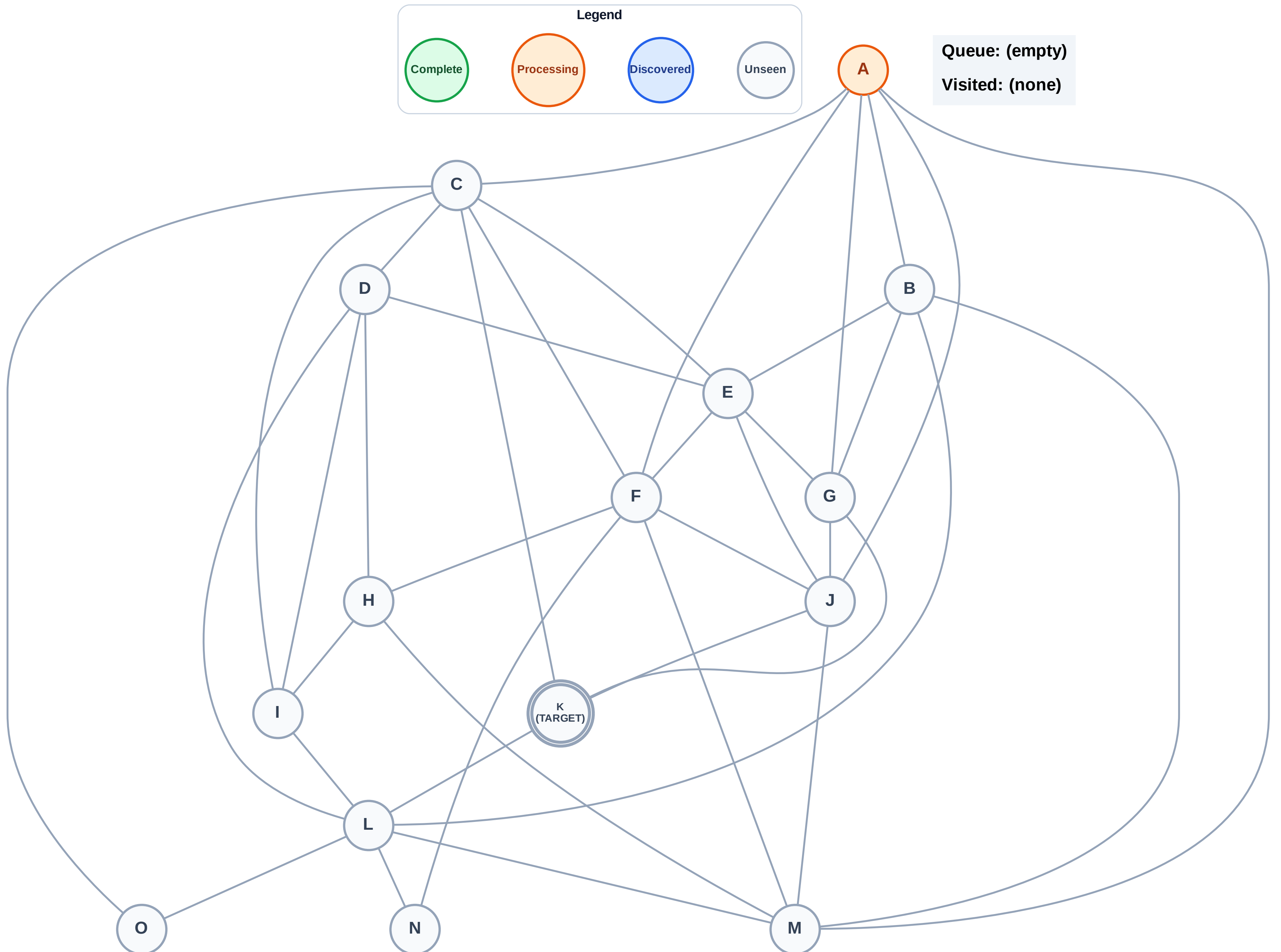
Discovered



Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

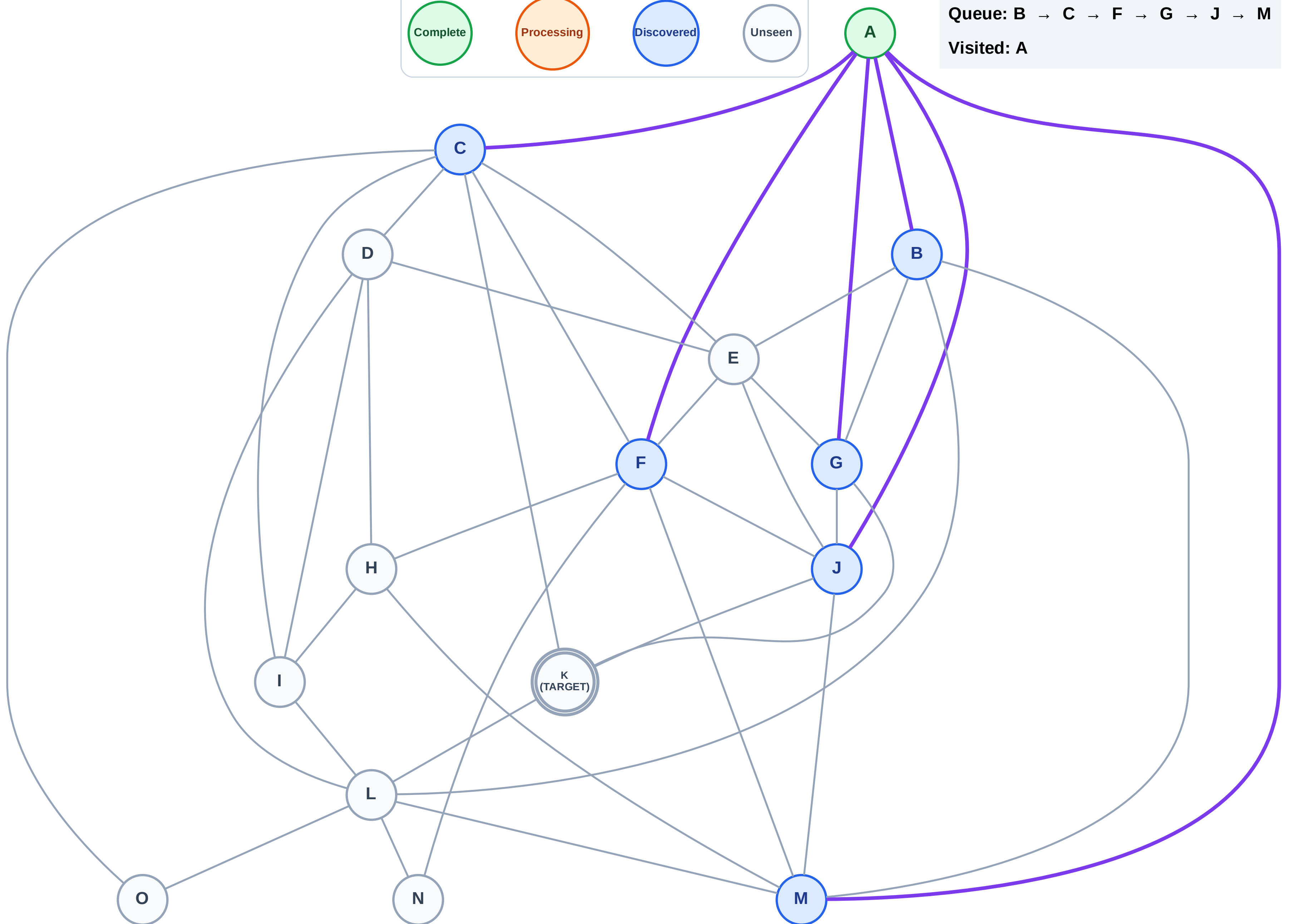
Step 3: Discover B, C, F, G, J, M from A

Legend



Queue: B → C → F → G → J → M

Visited: A



BFS Traversal

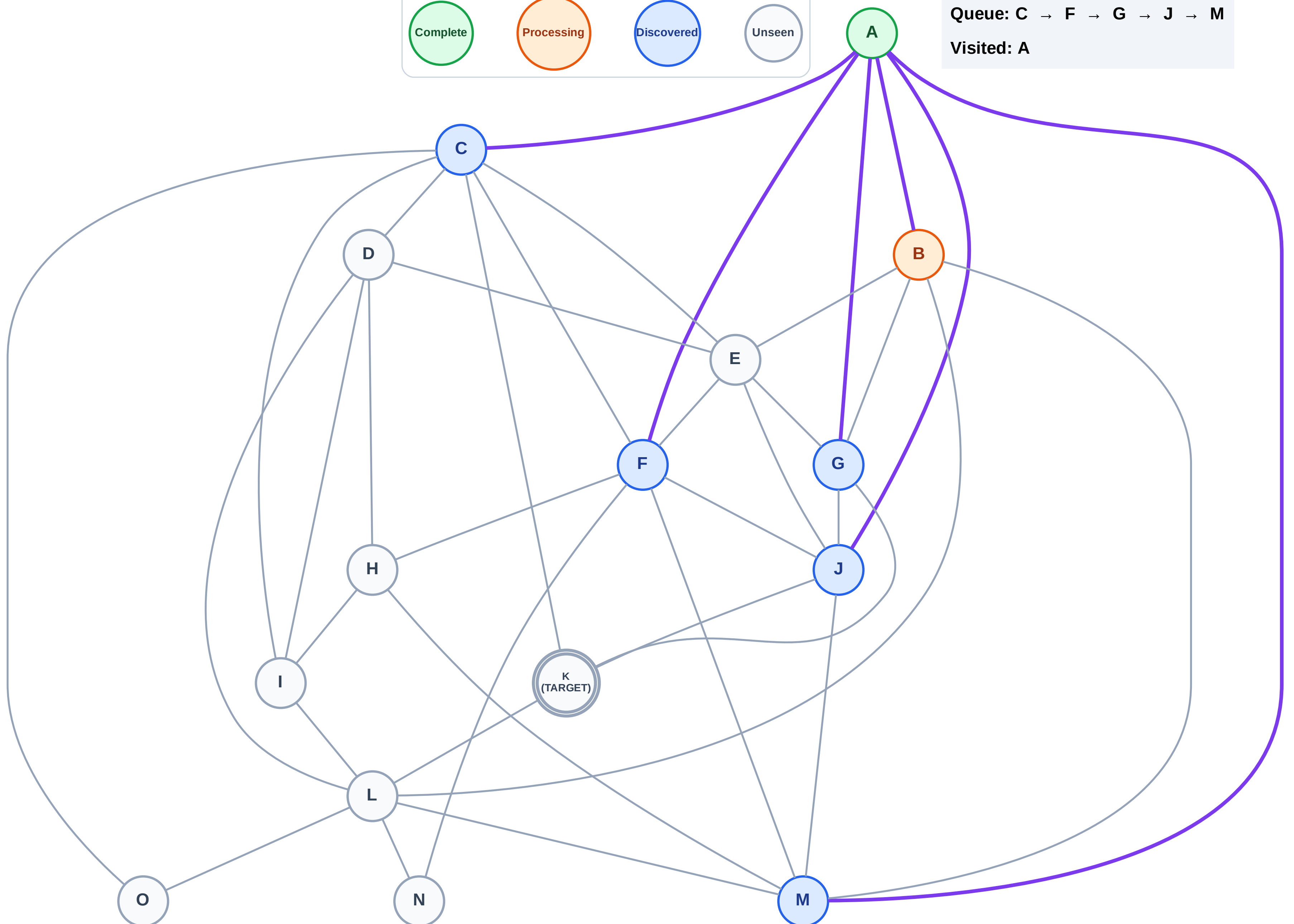
Step 4: Process node B

Legend



Queue: C → F → G → J → M

Visited: A



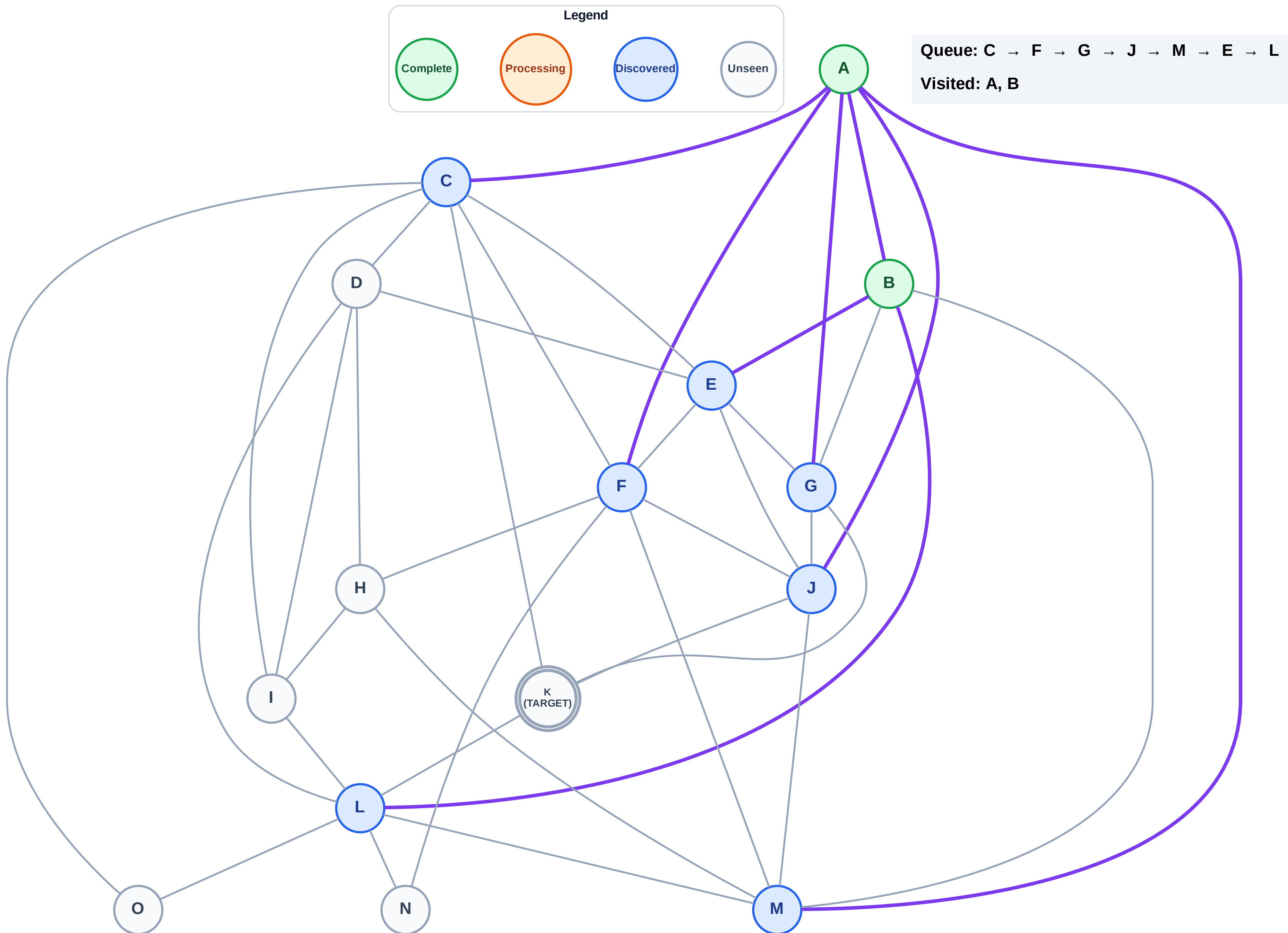
BFS Traversal

Step 5: Discover E, L from B

Legend



Queue: C → F → G → J → M → E → L
Visited: A, B



Step 6: Process node C

Step 6: Process node C



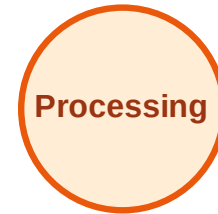
BFS Traversal

Step 7: Discover D, I, K, O from C

Legend



Complete



Processing



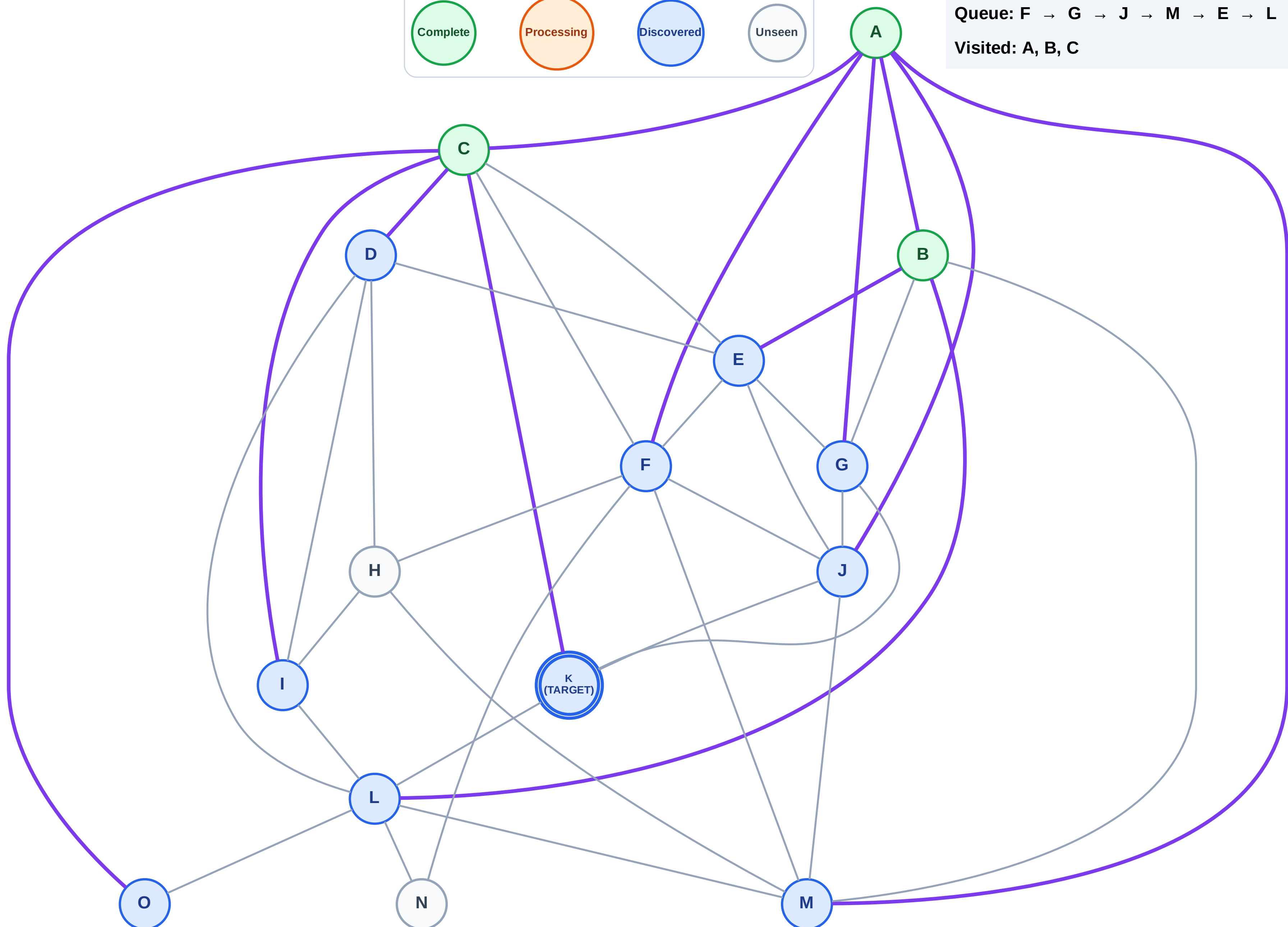
Discovered



Unseen

Queue: F → G → J → M → E → L → D → I → K → O

Visited: A, B, C



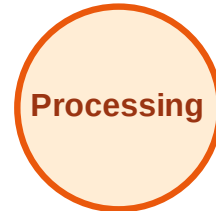
BFS Traversal

Step 8: Process node F

Legend



Complete



Processing



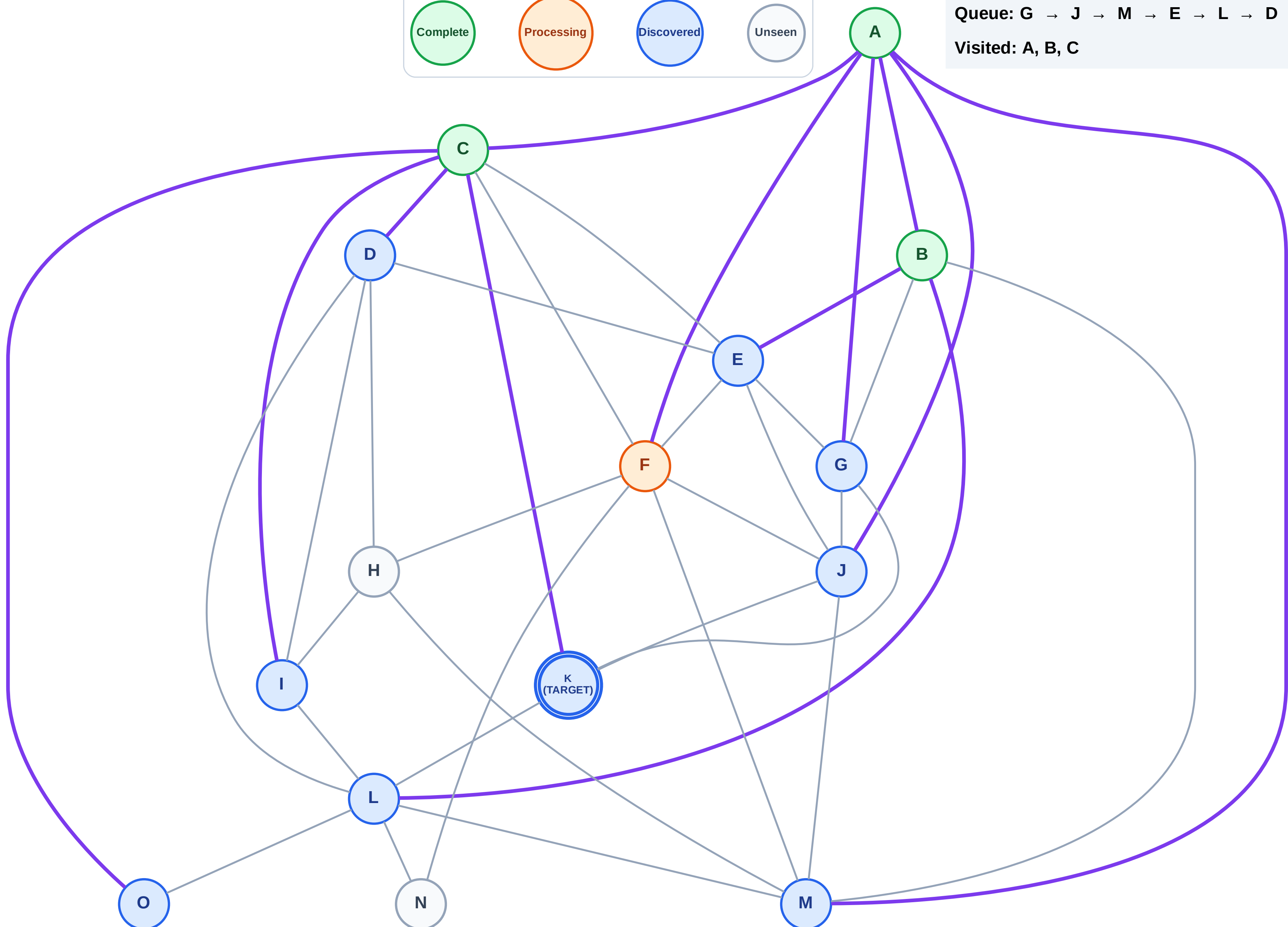
Discovered



Unseen

Queue: G → J → M → E → L → D → I → K → O

Visited: A, B, C



BFS Traversal

Step 9: Discover H, N from F

Legend



Complete



Processing



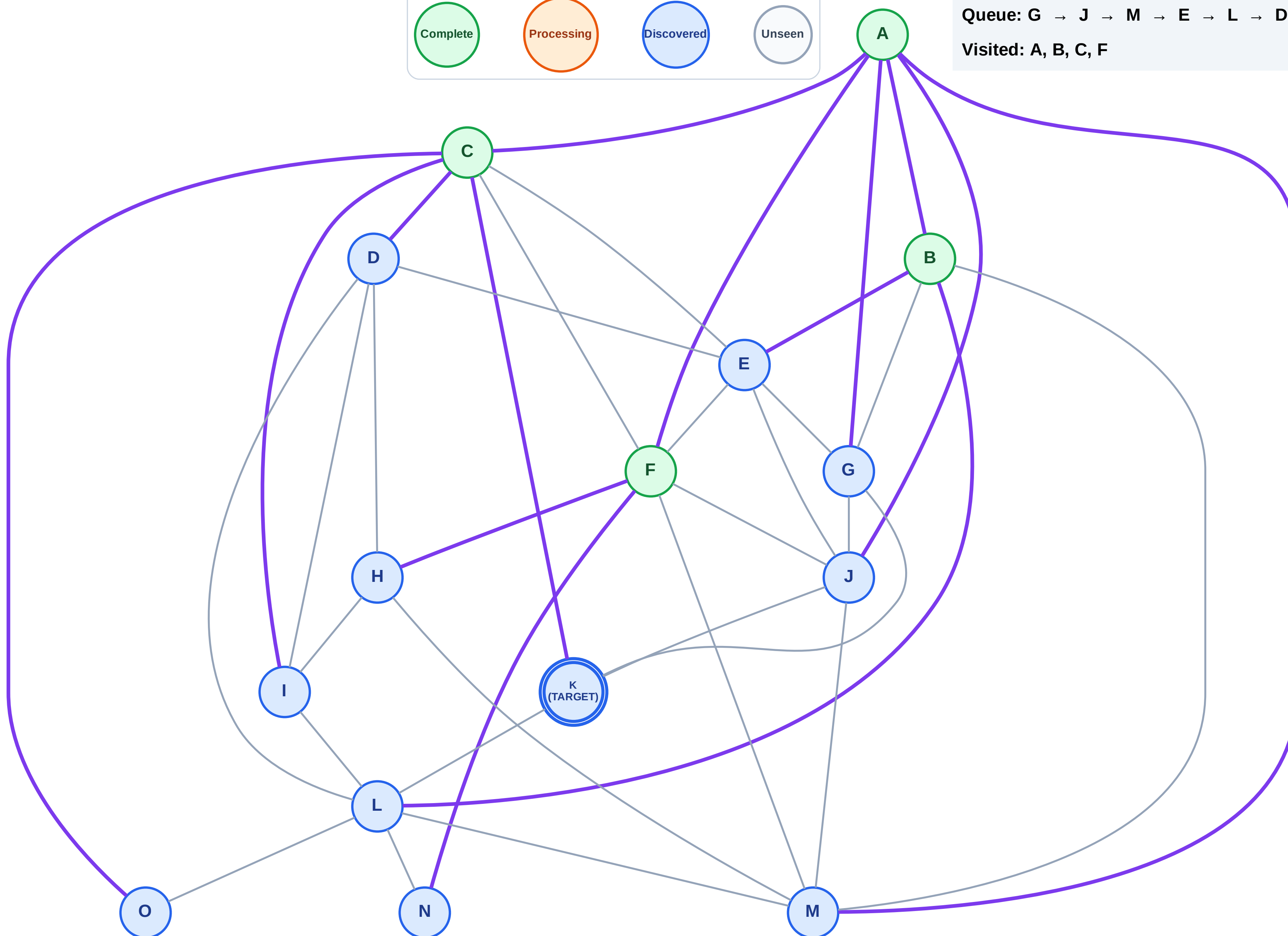
Discovered



Unseen

Queue: G → J → M → E → L → D → I → K → O → H → N

Visited: A, B, C, F



BFS Traversal

Step 10: Process node G

Legend

Complete

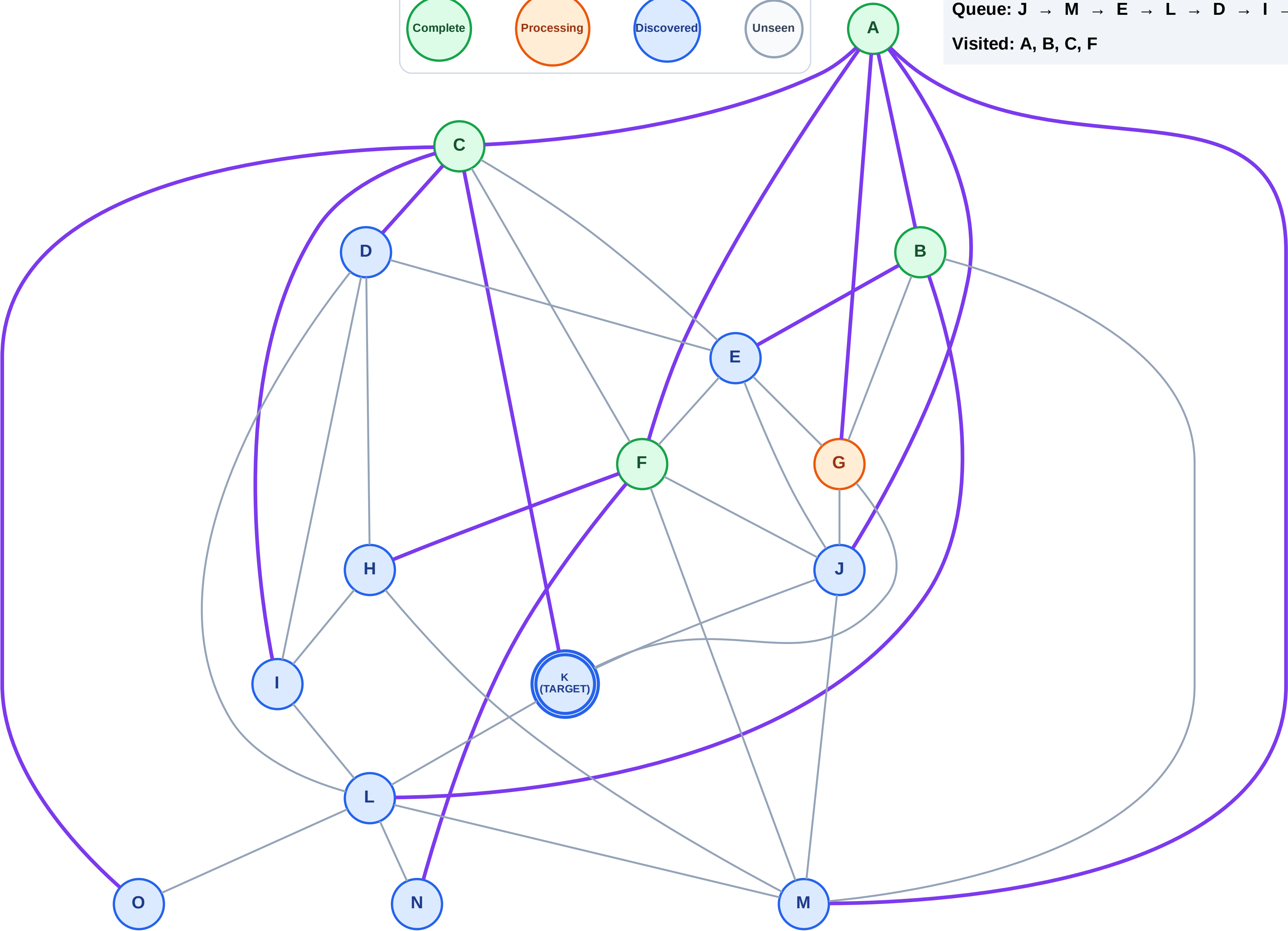
Processing

Discovered

Unseen

Queue: J → M → E → L → D → I → K → O → H → N

Visited: A, B, C, F



BFS Traversal

Step 11: No new neighbors from G

Legend



Complete



Processing



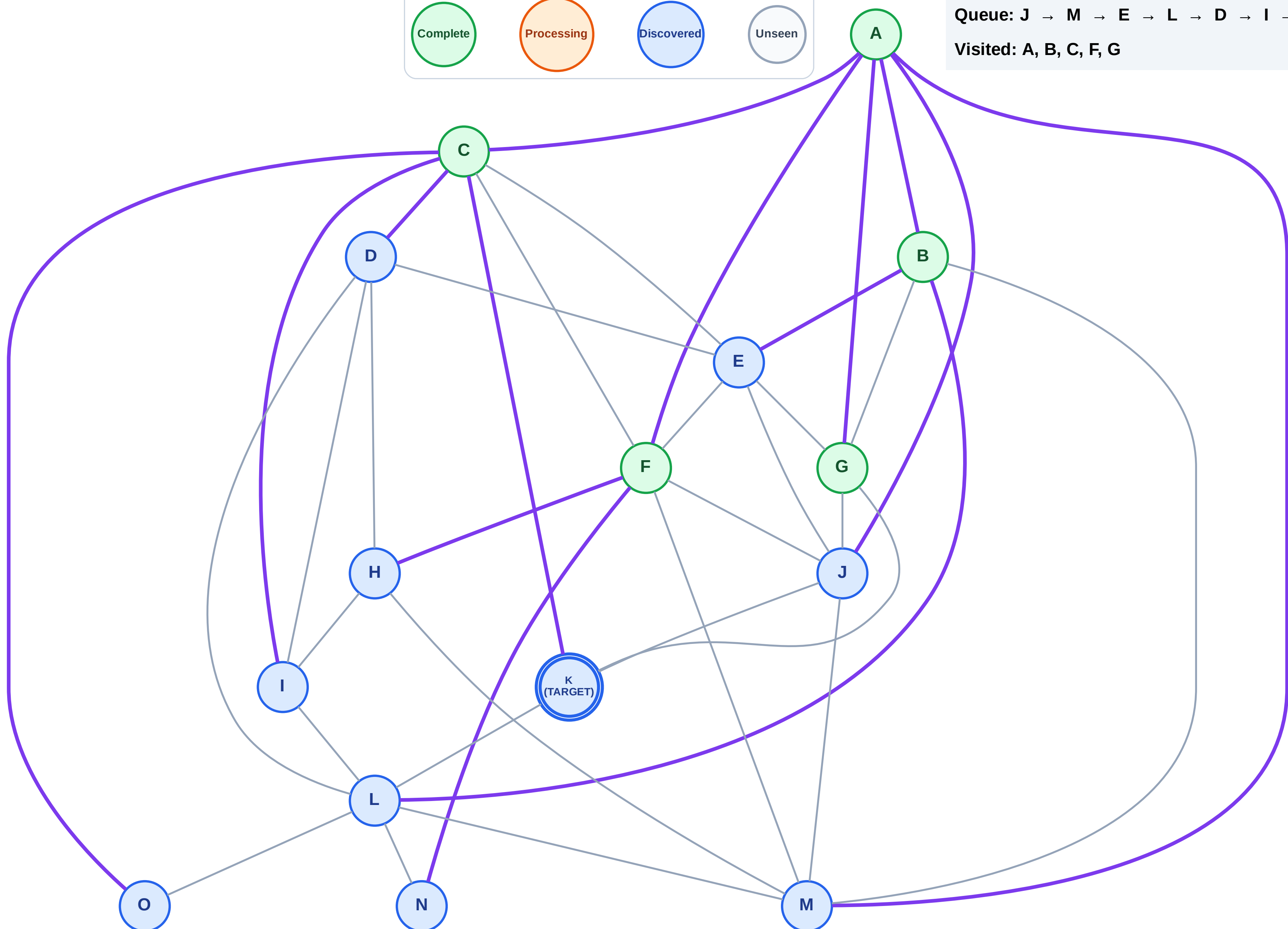
Discovered



Unseen

Queue: J → M → E → L → D → I → K → O → H → N

Visited: A, B, C, F, G



BFS Traversal

Step 12: Process node J

Legend

Complete

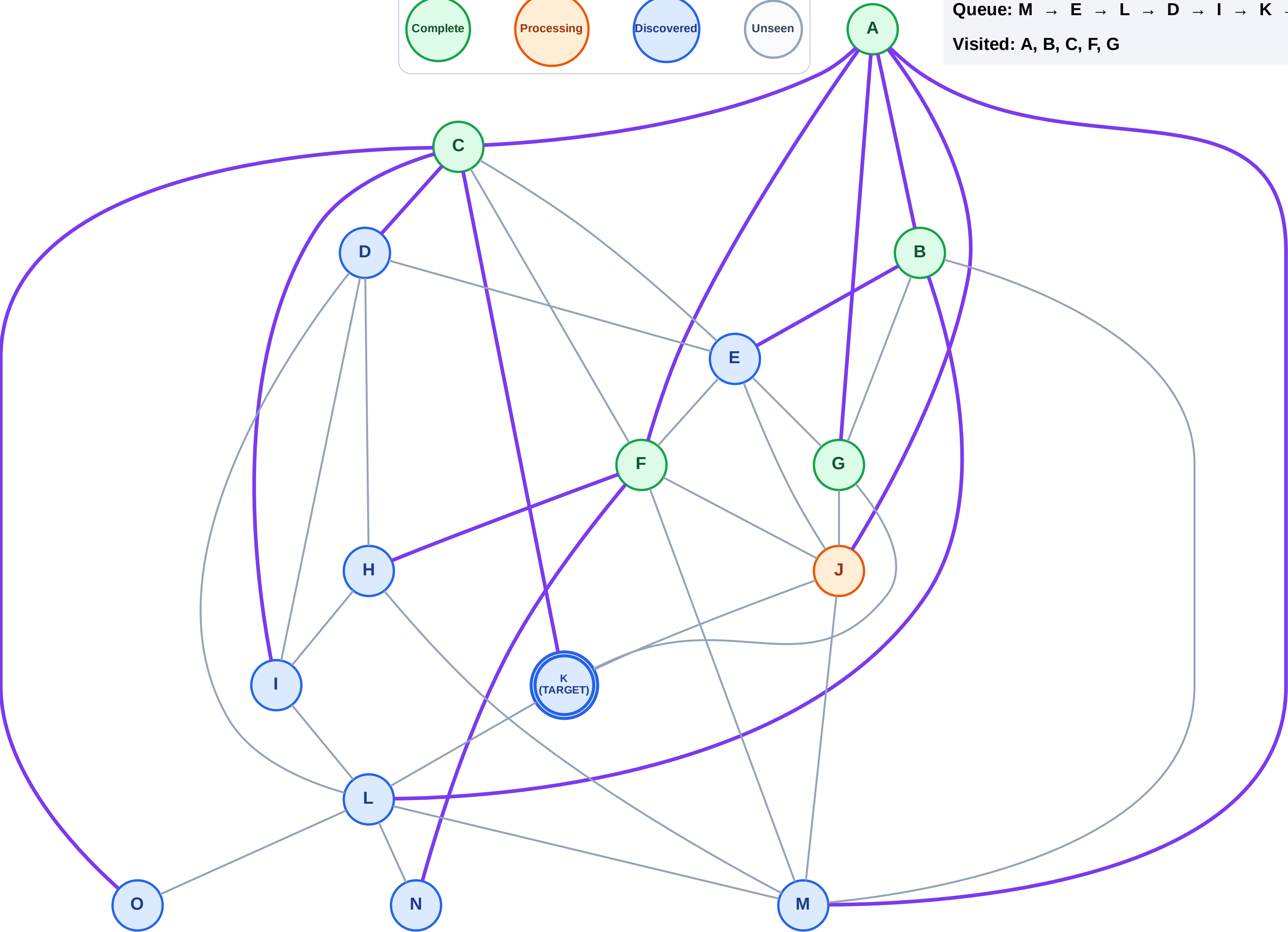
Processing

Discovered

Unseen

Queue: M → E → L → D → I → K → O → H → N

Visited: A, B, C, F, G



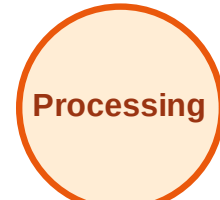
BFS Traversal

Step 13: No new neighbors from J

Legend



Complete



Processing



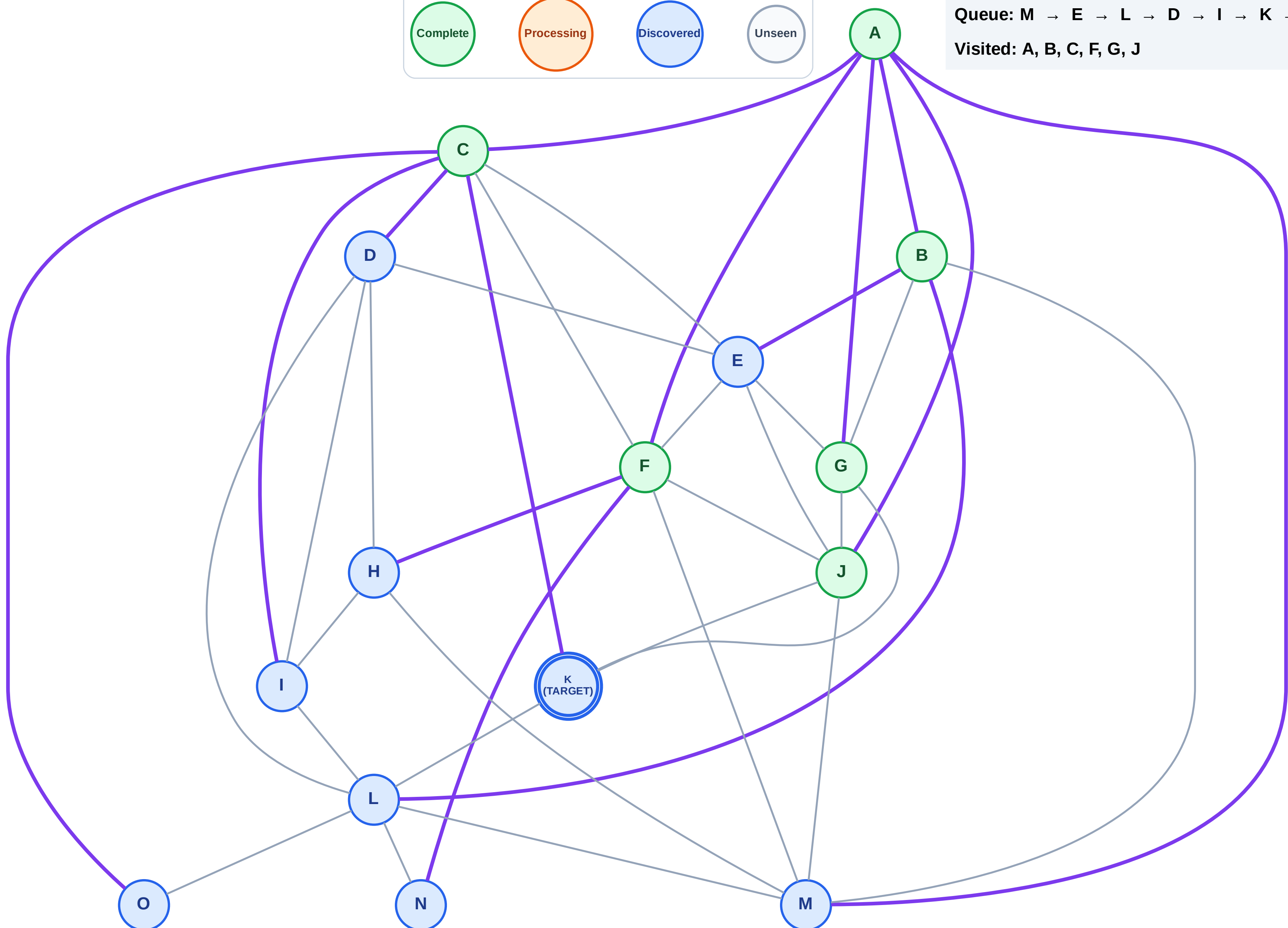
Discovered



Unseen

Queue: M → E → L → D → I → K → O → H → N

Visited: A, B, C, F, G, J



BFS Traversal

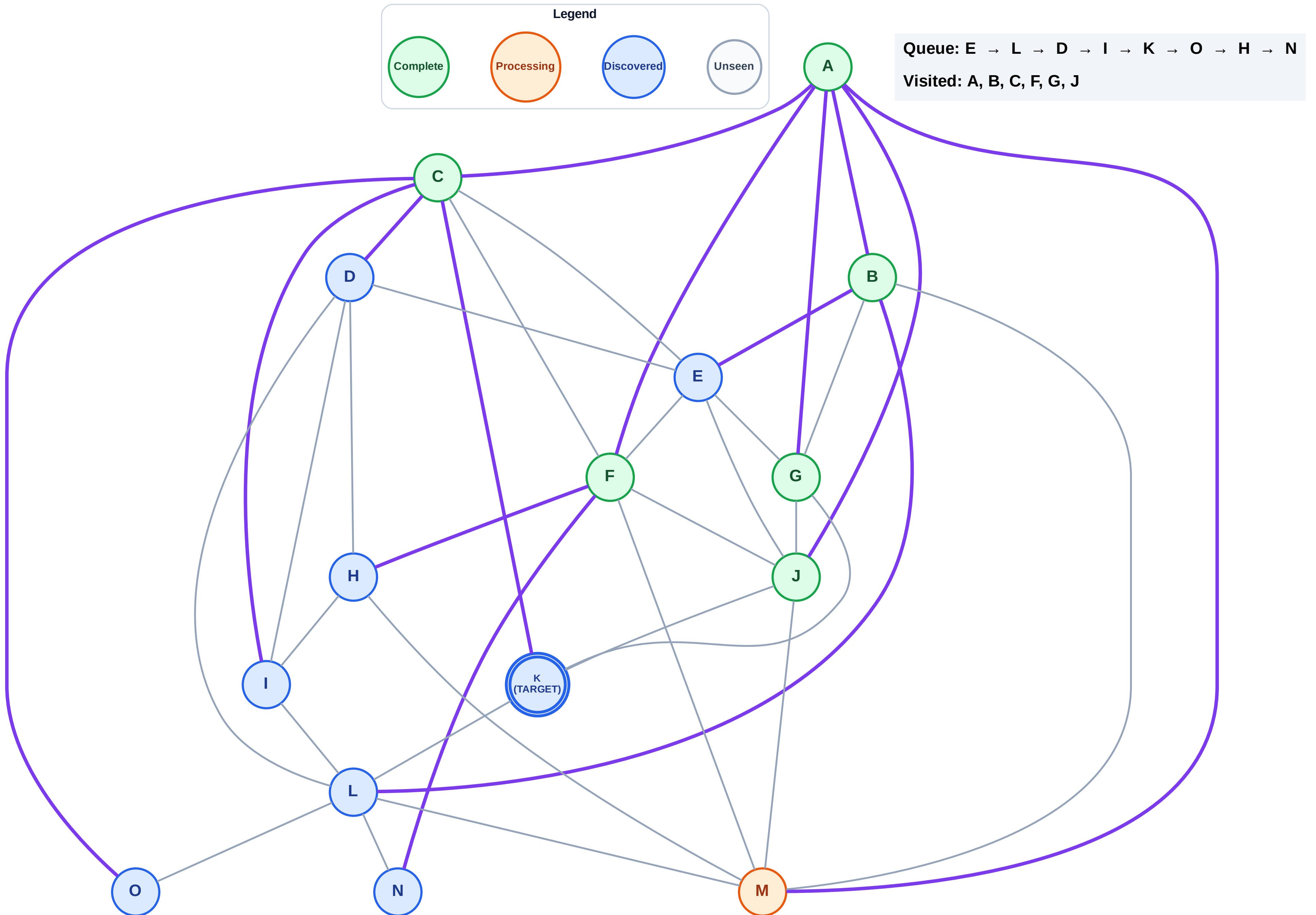
Step 14: Process node M

Legend



Queue: E → L → D → I → K → O → H → N

Visited: A, B, C, F, G, J

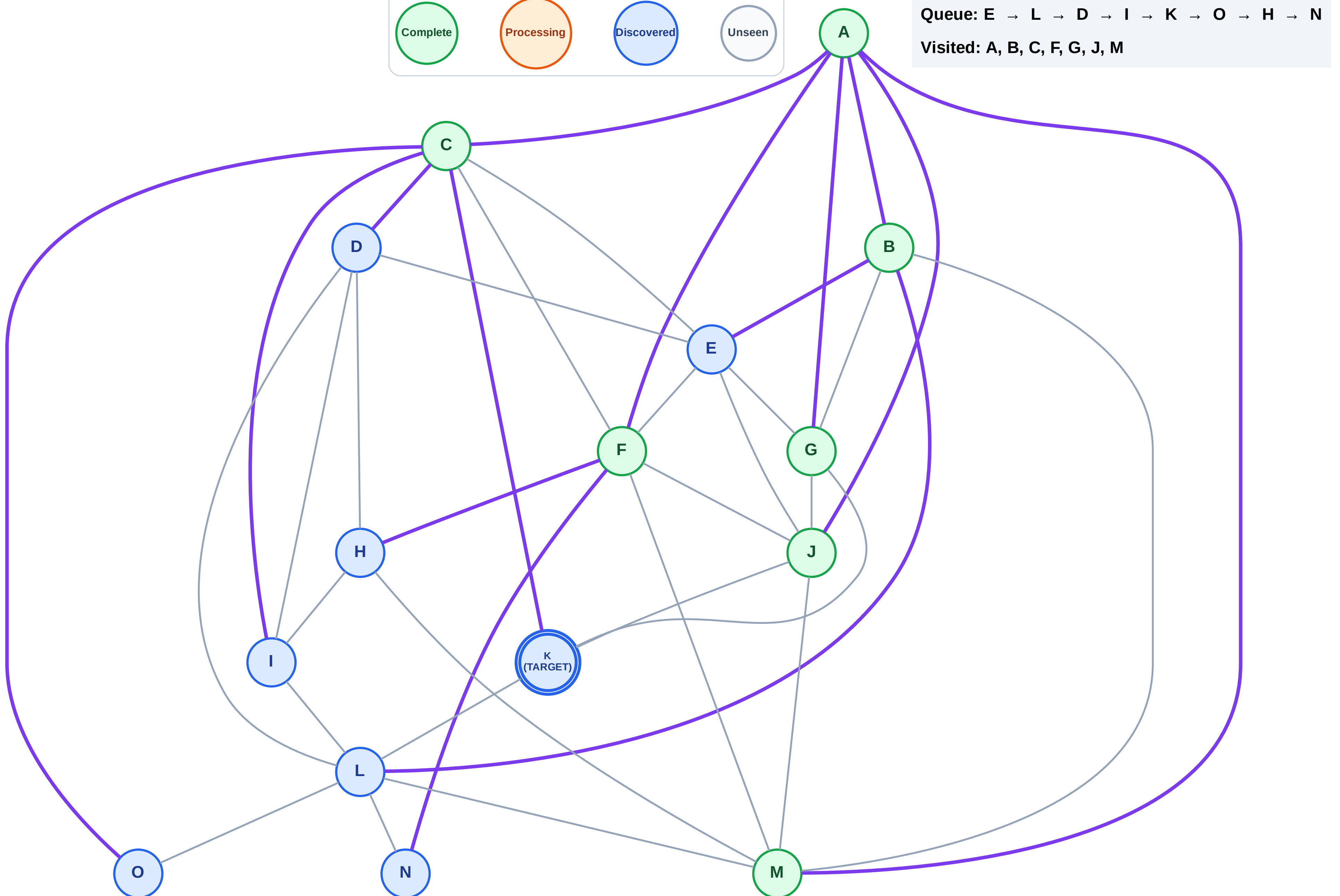


Step 15: No new neighbors from M

Step 15: No new neighbors from M

Complete Processing Discovered Unseen

Visited: A, B, C, F, G, J, M



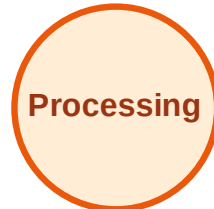
BFS Traversal

Step 16: Process node E

Legend



Complete



Processing

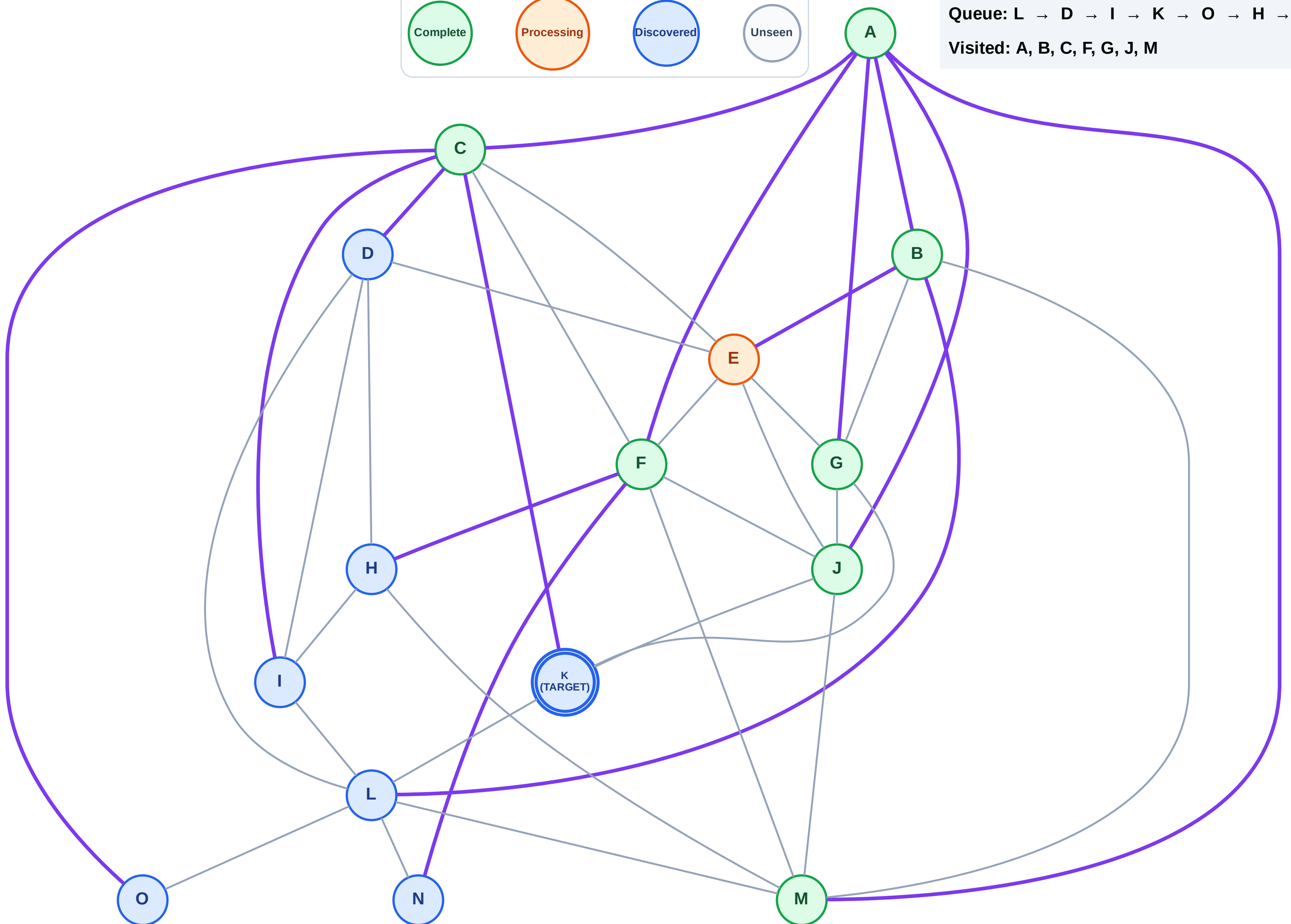


Discovered



Unseen

Queue: L → D → I → K → O → H → N
Visited: A, B, C, F, G, J, M



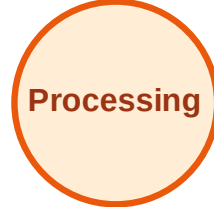
BFS Traversal

Step 17: No new neighbors from E

Legend



Complete



Processing



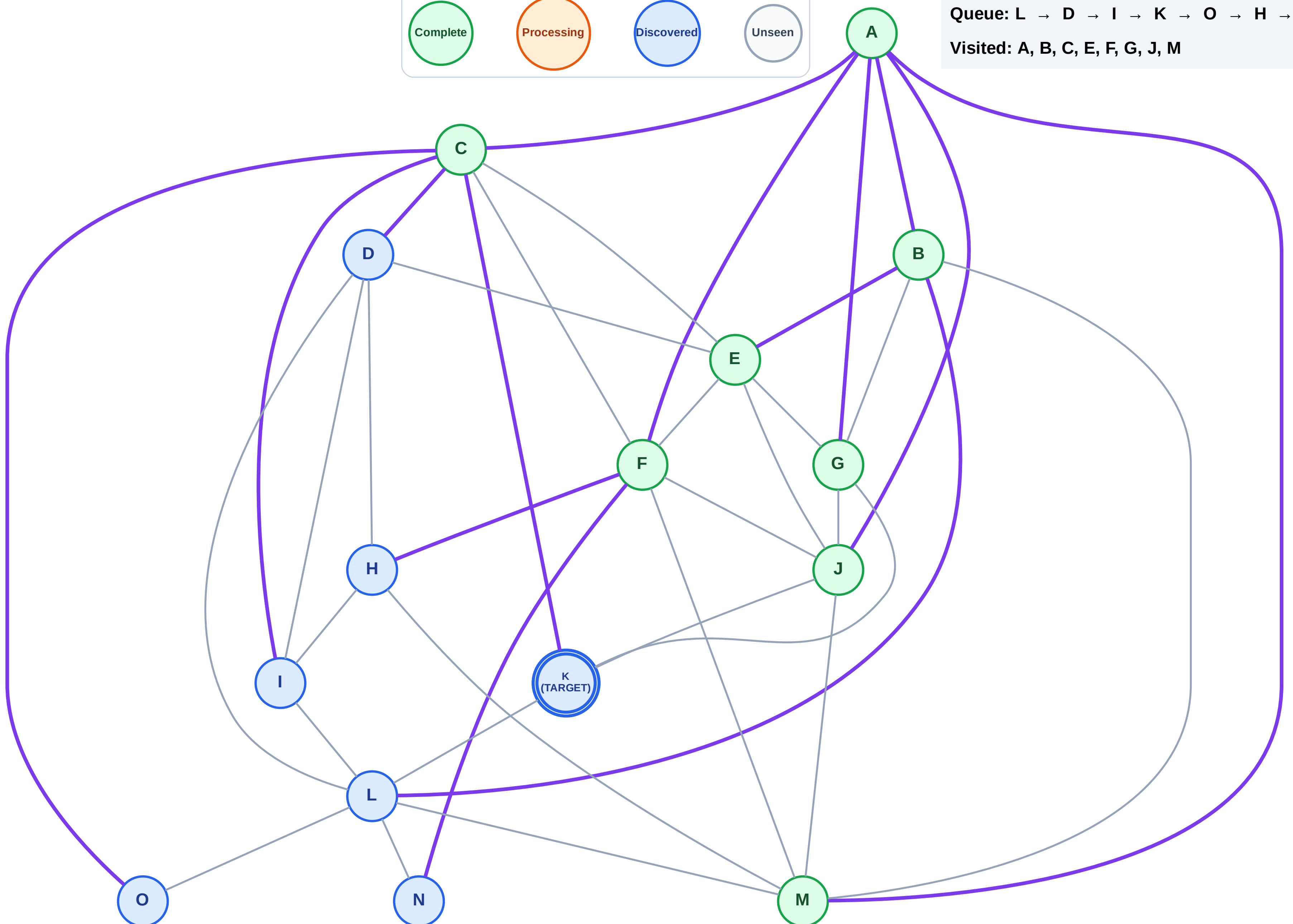
Discovered



Unseen

Queue: L → D → I → K → O → H → N

Visited: A, B, C, E, F, G, J, M



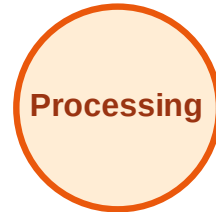
BFS Traversal

Step 18: Process node L

Legend



Complete



Processing



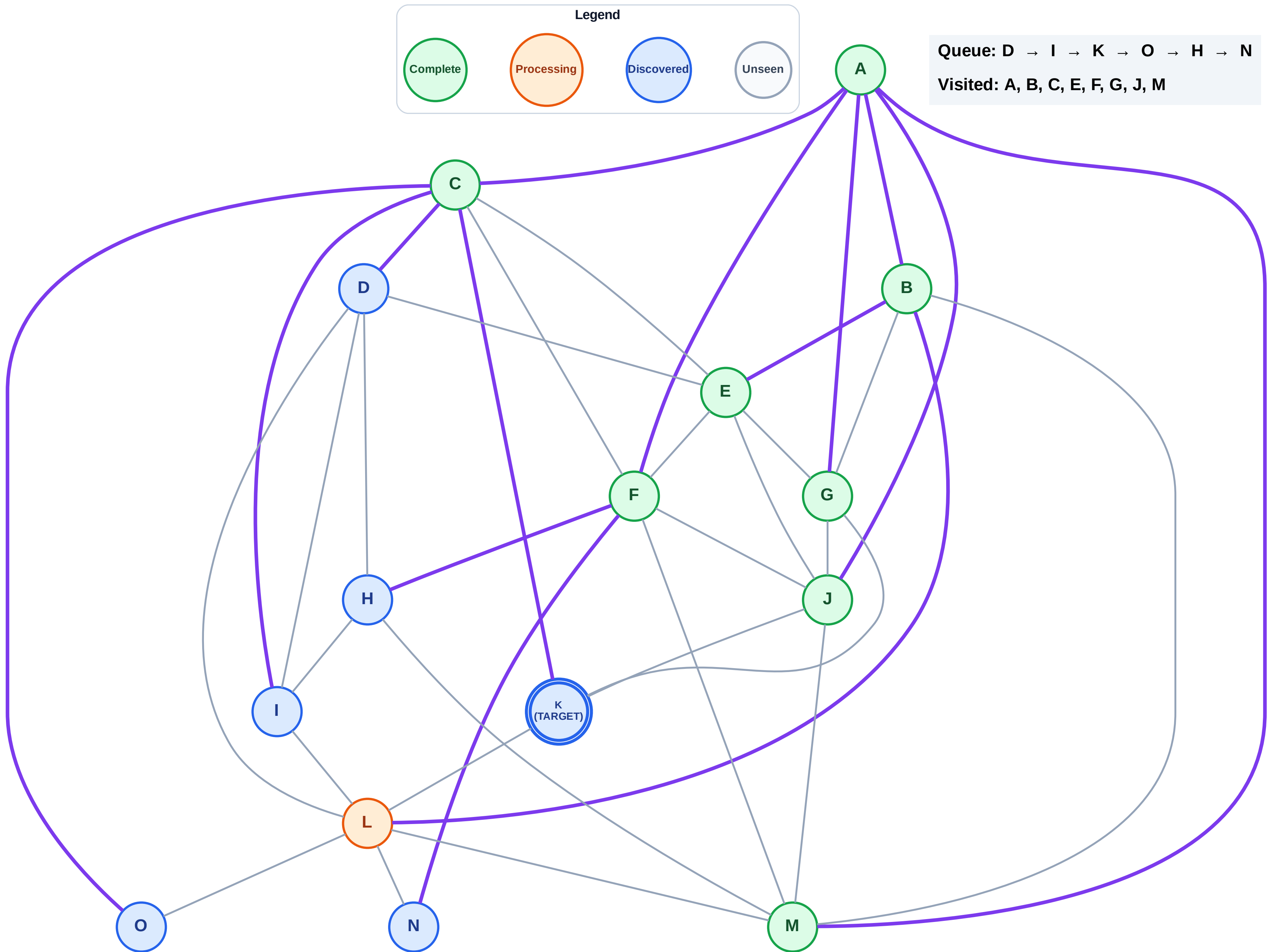
Discovered



Unseen

Queue: D → I → K → O → H → N

Visited: A, B, C, E, F, G, J, M



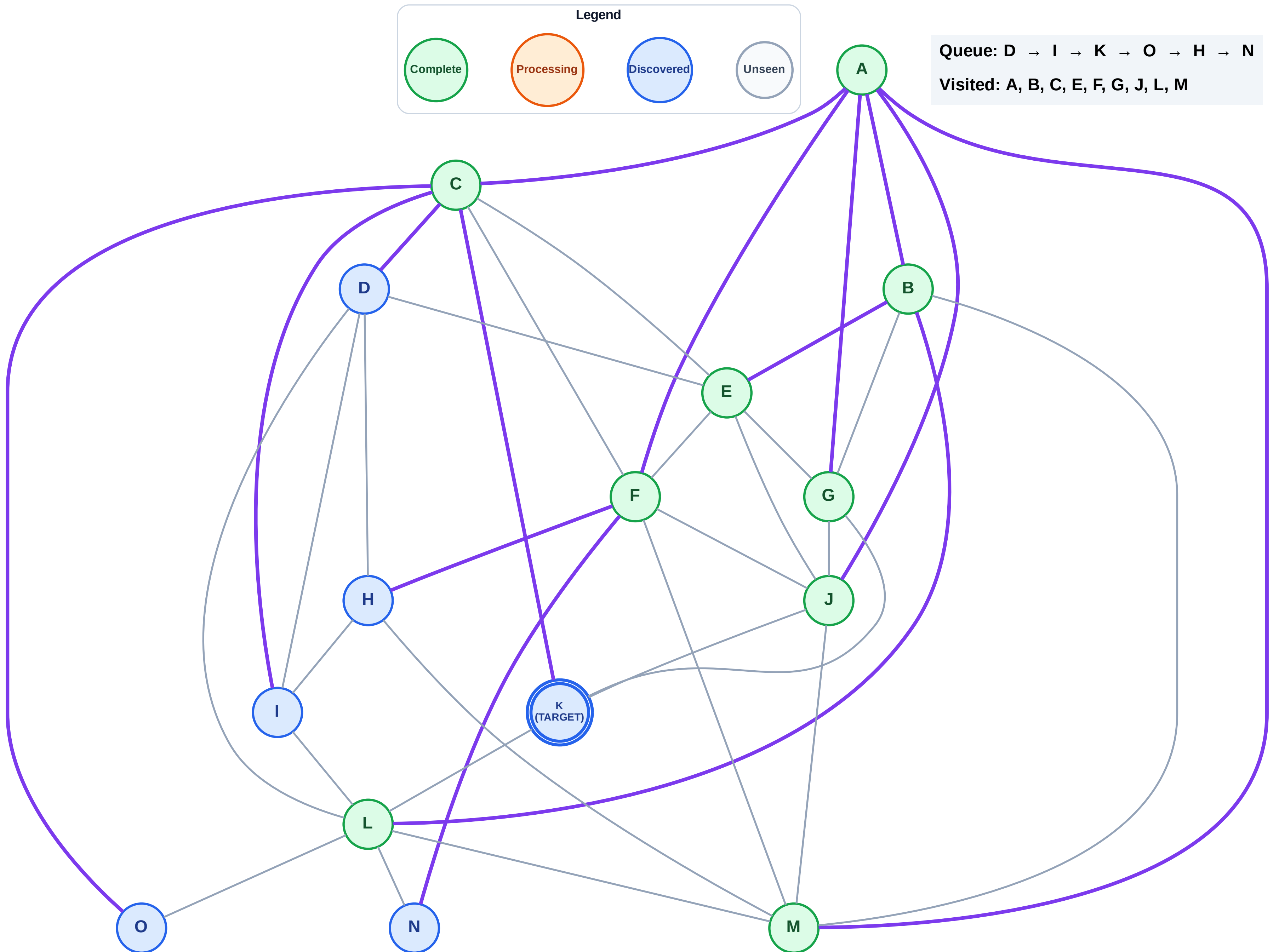
BFS Traversal

Step 19: No new neighbors from L

Legend



Queue: D → I → K → O → H → N
Visited: A, B, C, E, F, G, J, L, M



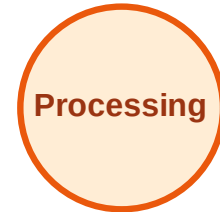
BFS Traversal

Step 20: Process node D

Legend



Complete



Processing



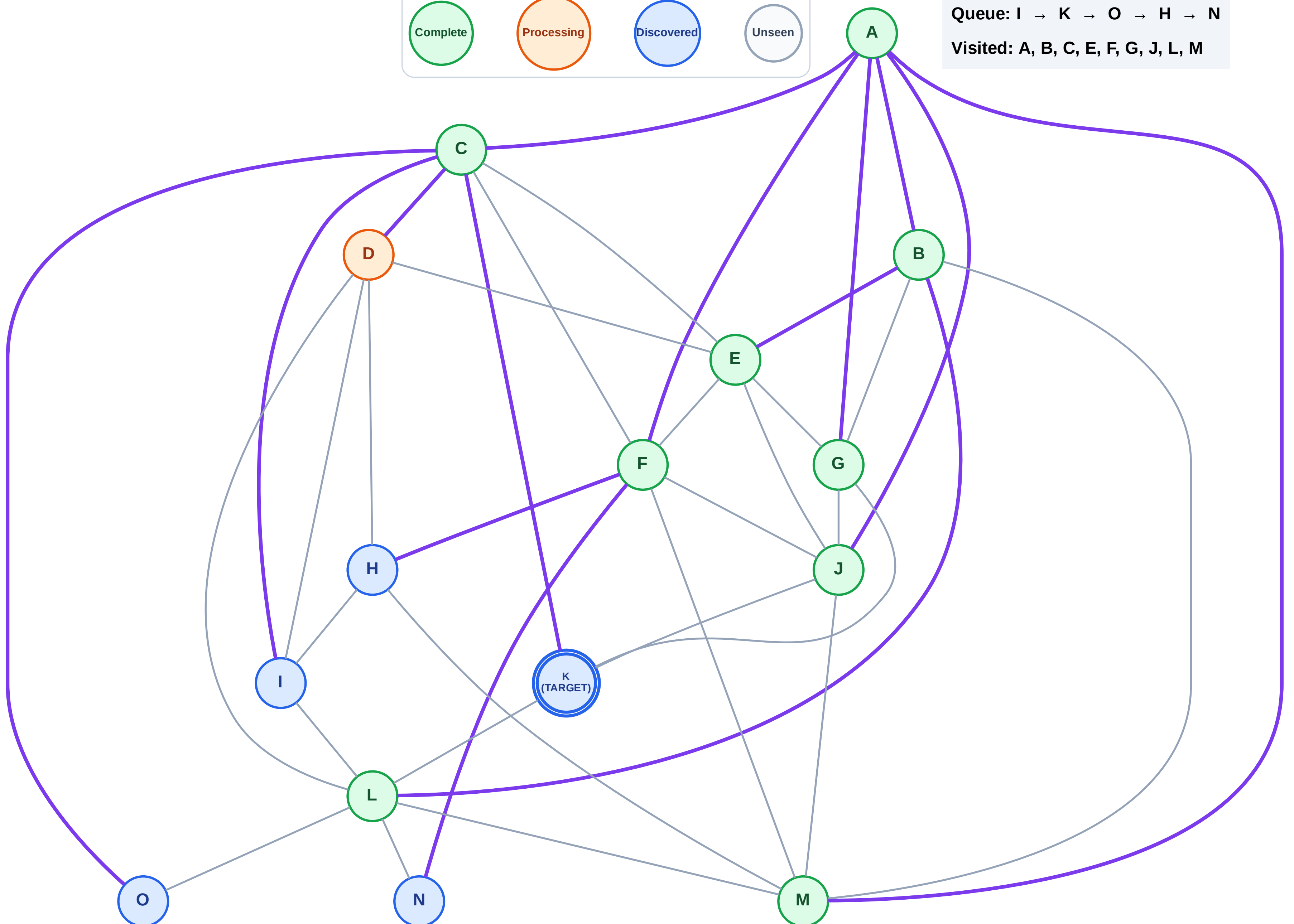
Discovered



Unseen

Queue: I → K → O → H → N

Visited: A, B, C, E, F, G, J, L, M



BFS Traversal

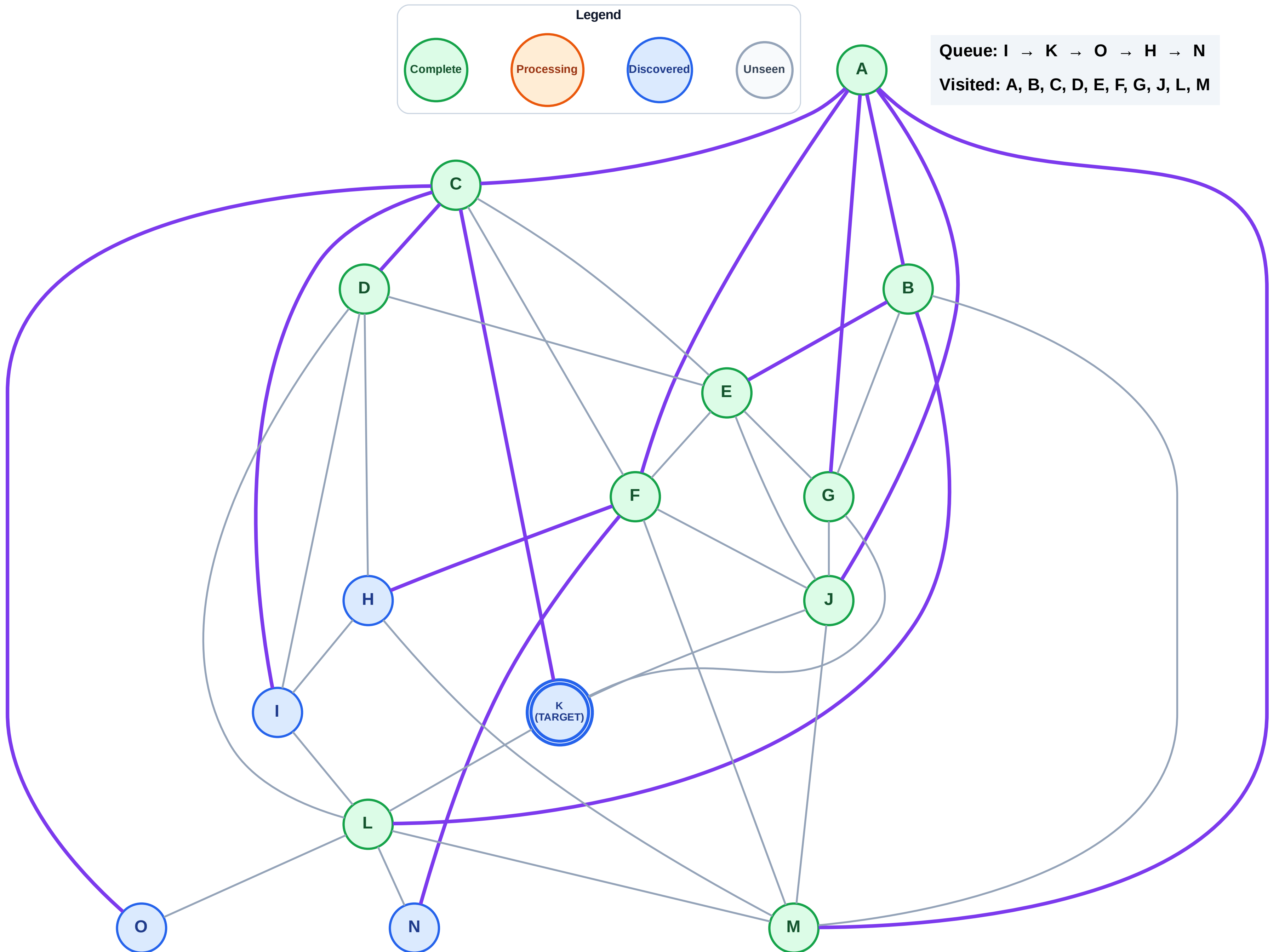
Step 21: No new neighbors from D

Legend



Queue: I → K → O → H → N

Visited: A, B, C, D, E, F, G, J, L, M



BFS Traversal

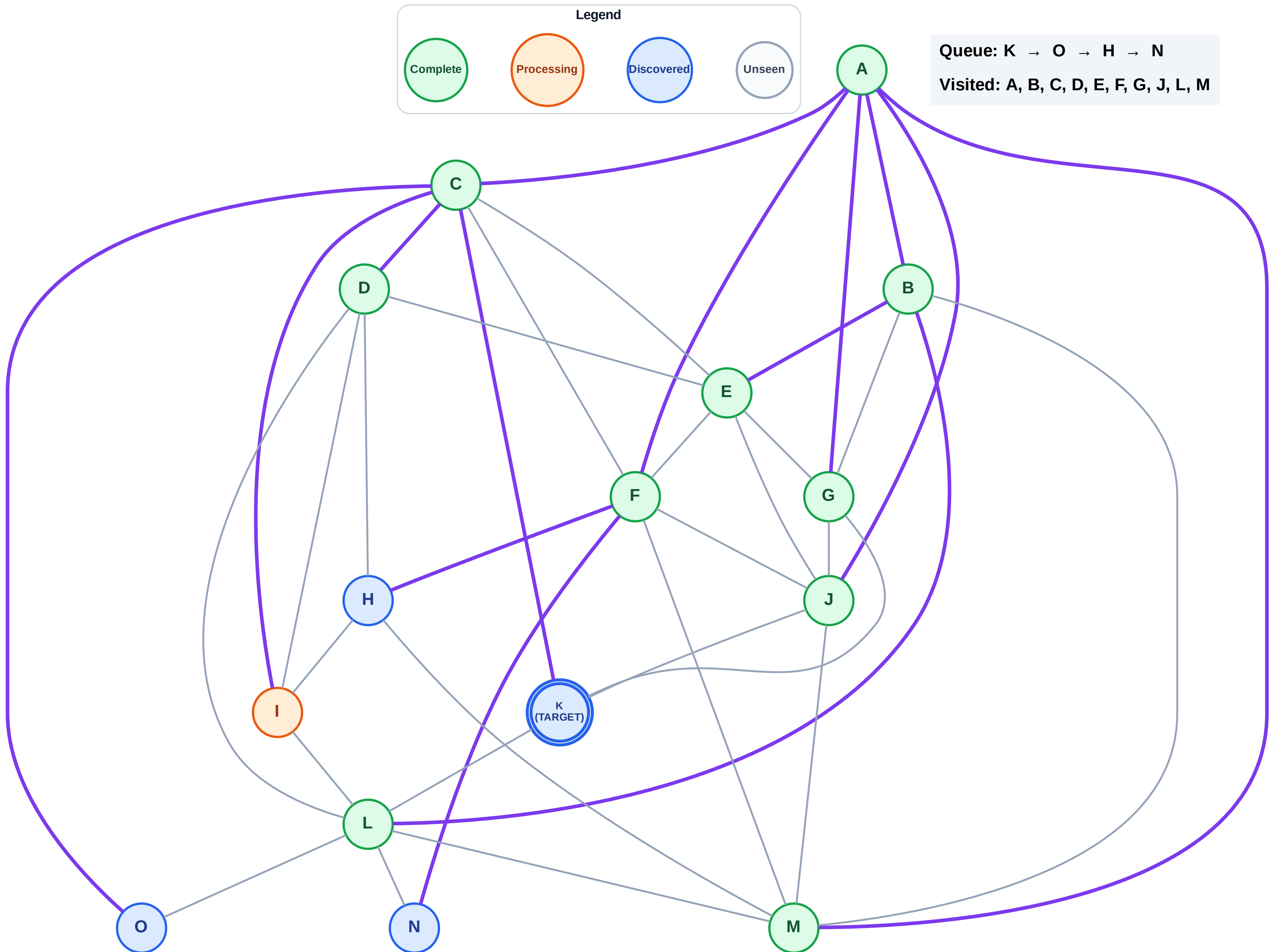
Step 22: Process node I

Legend



Queue: K → O → H → N

Visited: A, B, C, D, E, F, G, J, L, M



BFS Traversal

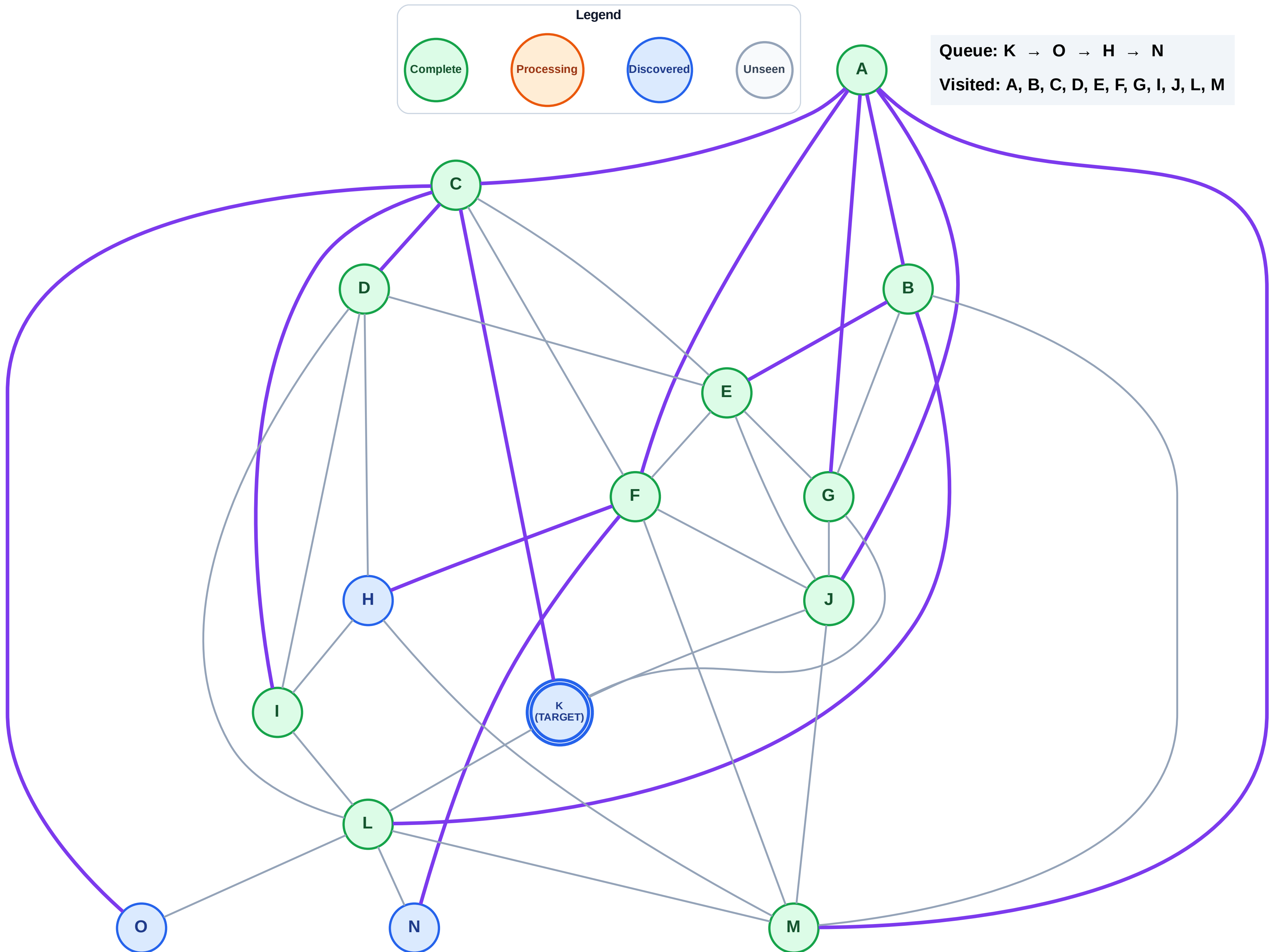
Step 23: No new neighbors from I

Legend



Queue: K → O → H → N

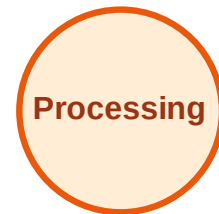
Visited: A, B, C, D, E, F, G, I, J, L, M



BFS Traversal

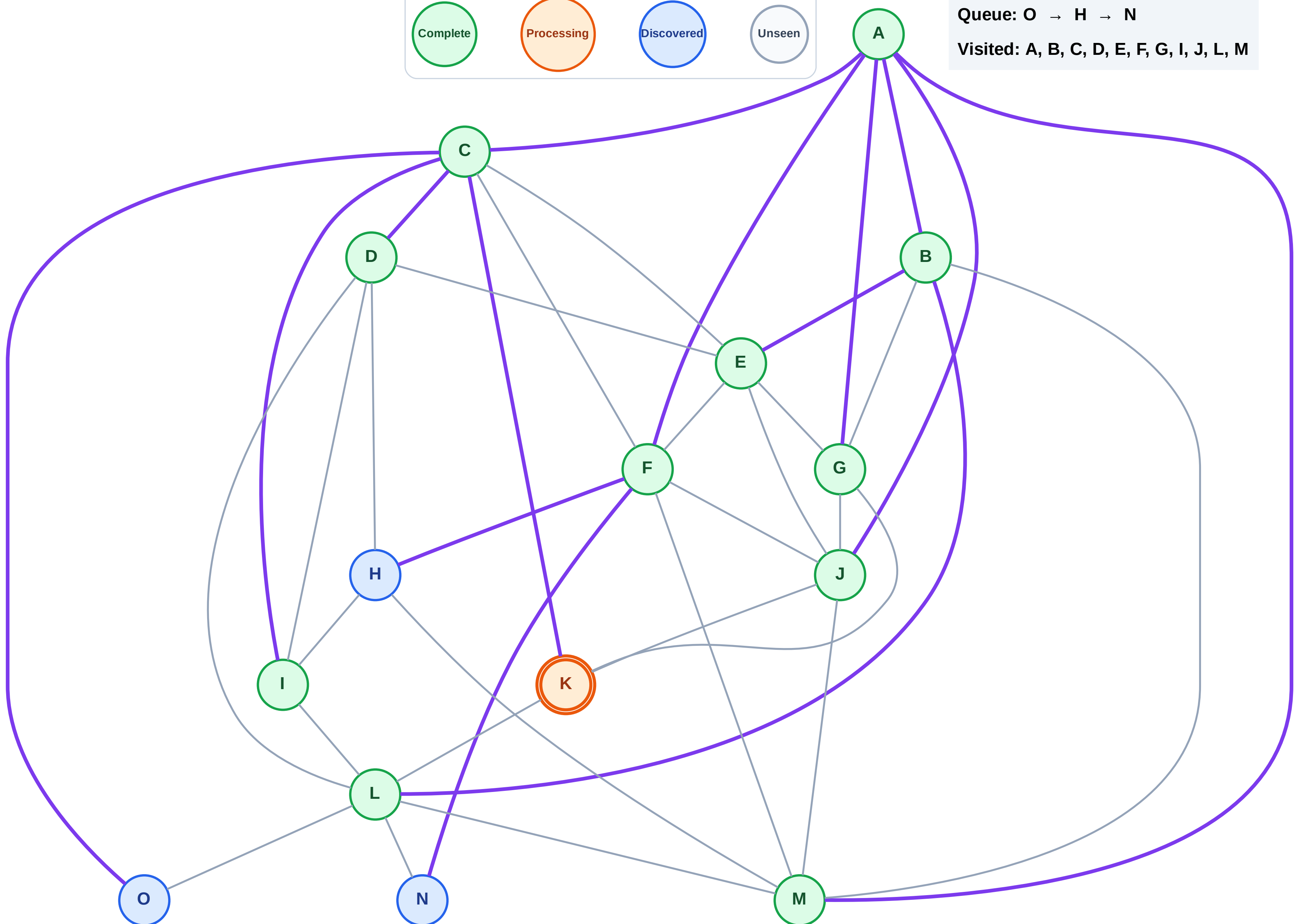
Step 24: Process node K

Legend



Queue: O → H → N

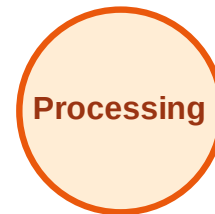
Visited: A, B, C, D, E, F, G, I, J, L, M



BFS Traversal

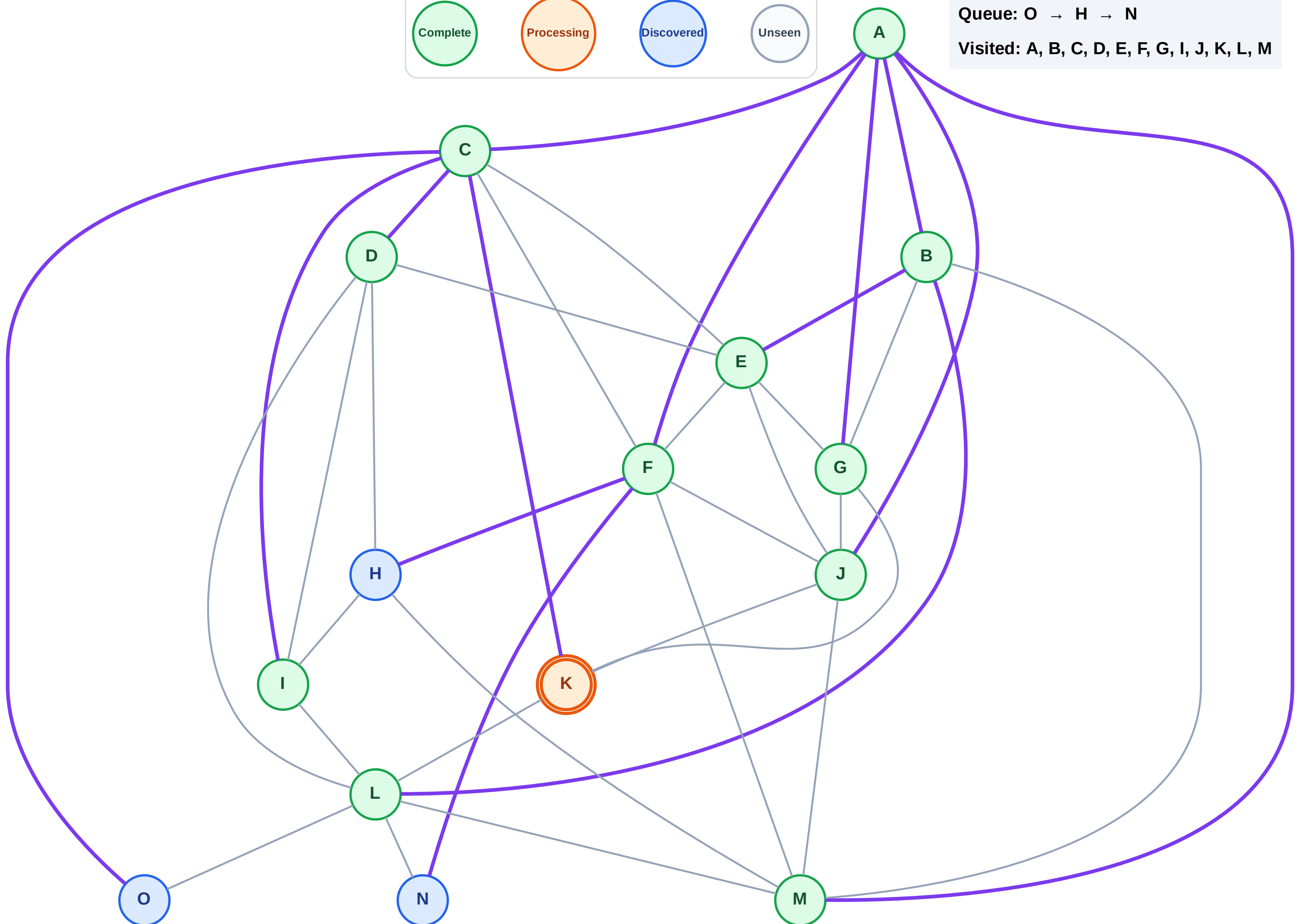
Step 25: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend



Queue: O → H → N

Visited: A, B, C, D, E, F, G, I, J, K, L, M



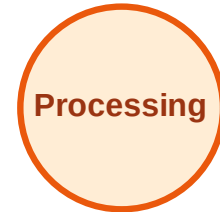
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing

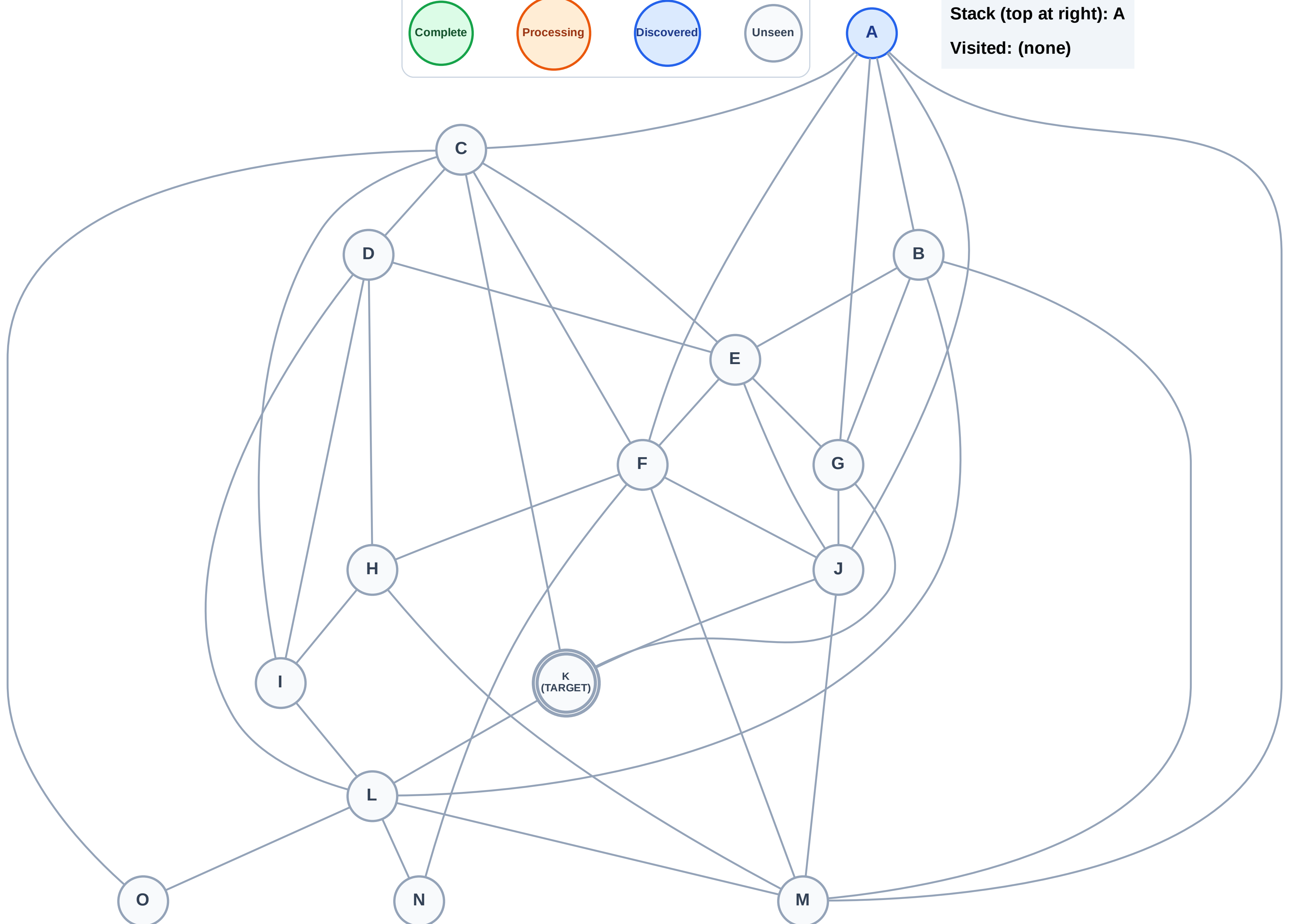


Discovered



Unseen

Stack (top at right): A
Visited: (none)



Step 2: Process node A

Step 2: Process node A

Complete



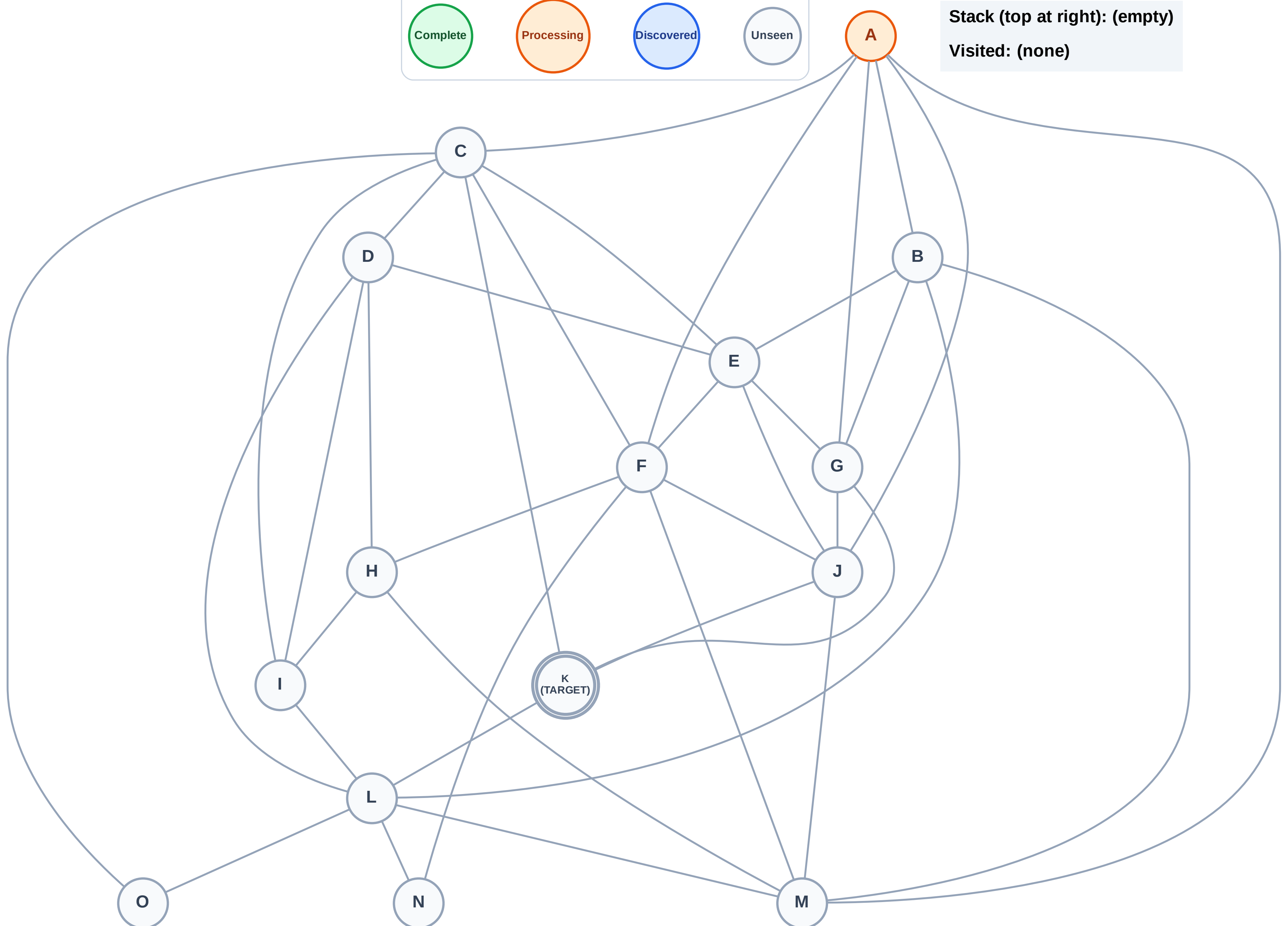
Processing



Discovered

Unseen

Visited: (none)



DFS Traversal

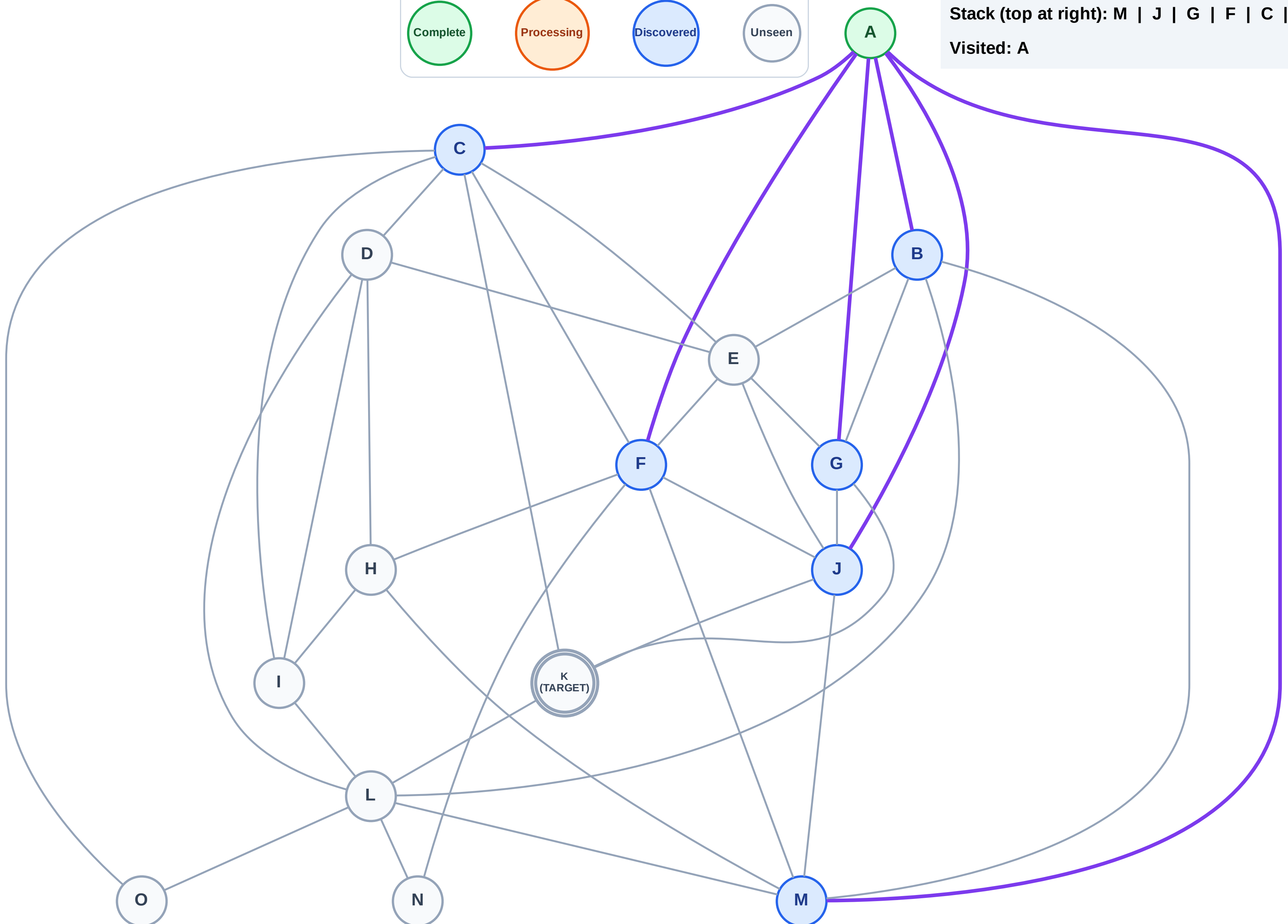
Step 3: Discover M, J, G, F, C, B from A

Legend



Stack (top at right): M | J | G | F | C | B

Visited: A



DFS Traversal

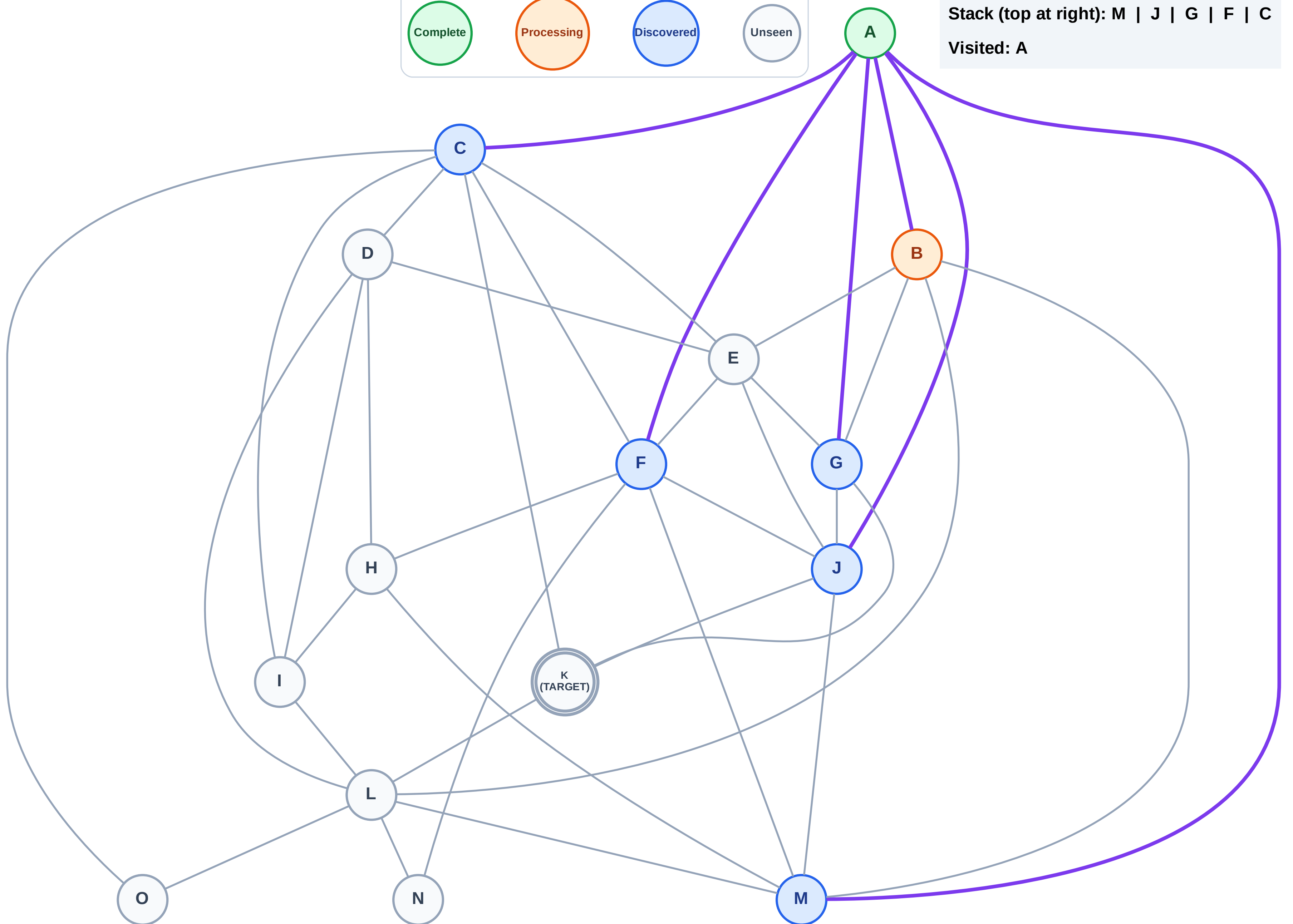
Step 4: Process node B

Legend



Stack (top at right): M | J | G | F | C

Visited: A



DFS Traversal

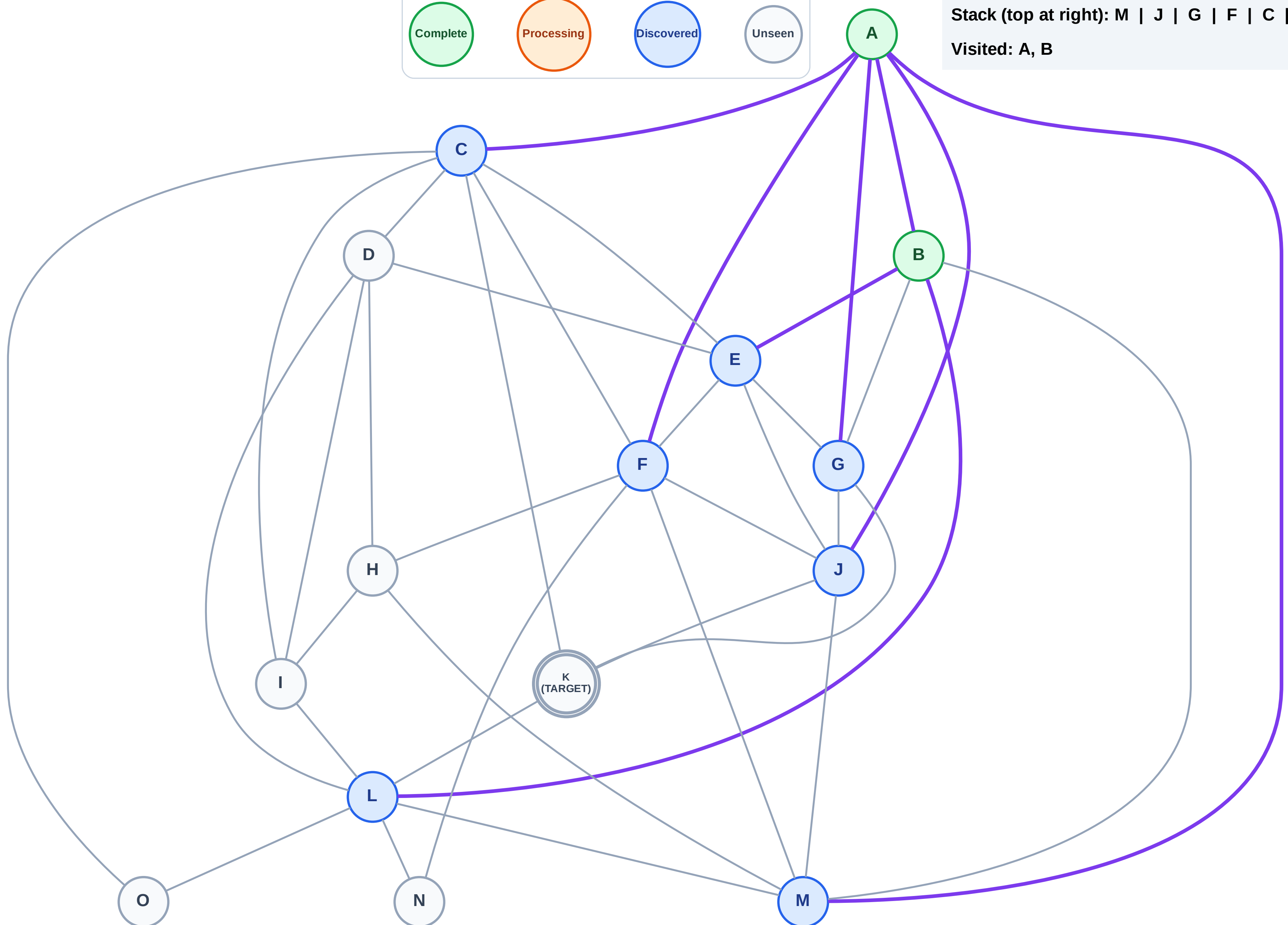
Step 5: Discover L, E from B

Legend



Stack (top at right): M | J | G | F | C | L | E

Visited: A, B



DFS Traversal

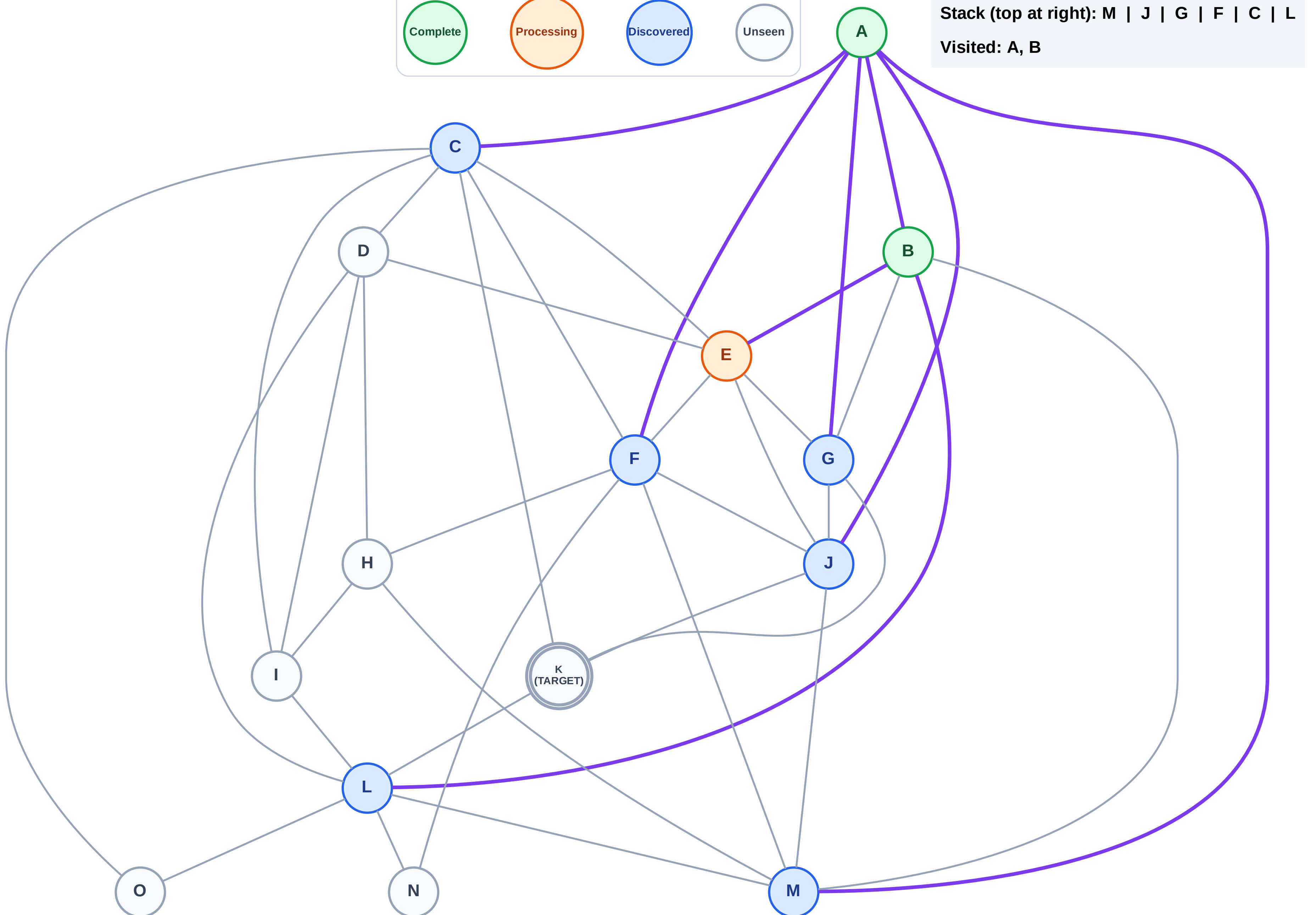
Step 6: Process node E

Legend



Stack (top at right): M | J | G | F | C | L

Visited: A, B



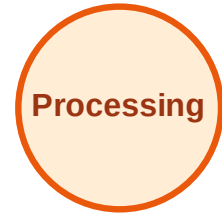
DFS Traversal

Step 7: Discover D from E

Legend



Complete



Processing



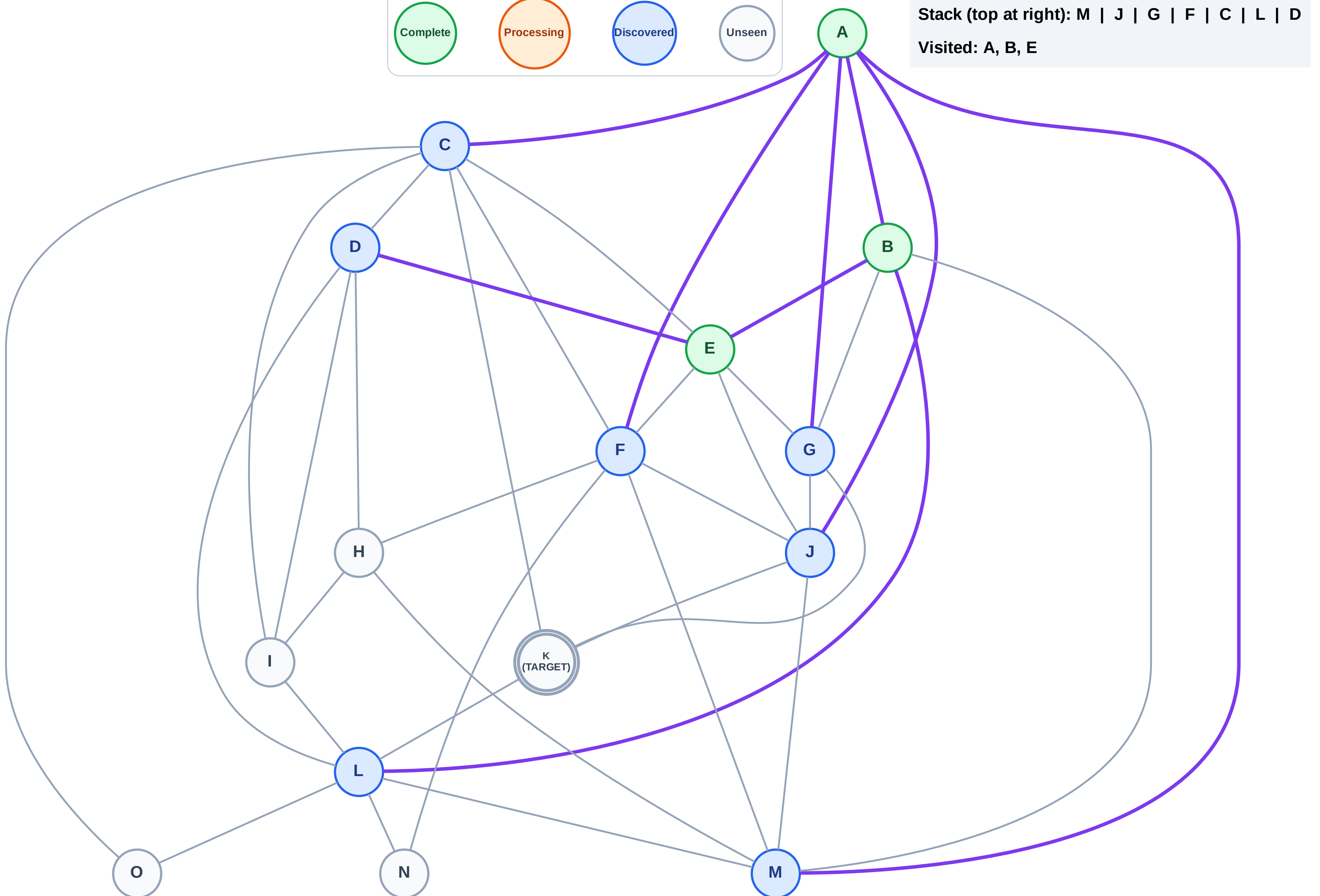
Discovered



Unseen

Stack (top at right): M | J | G | F | C | L | D

Visited: A, B, E



DFS Traversal

Step 8: Process node D

Legend

Complete

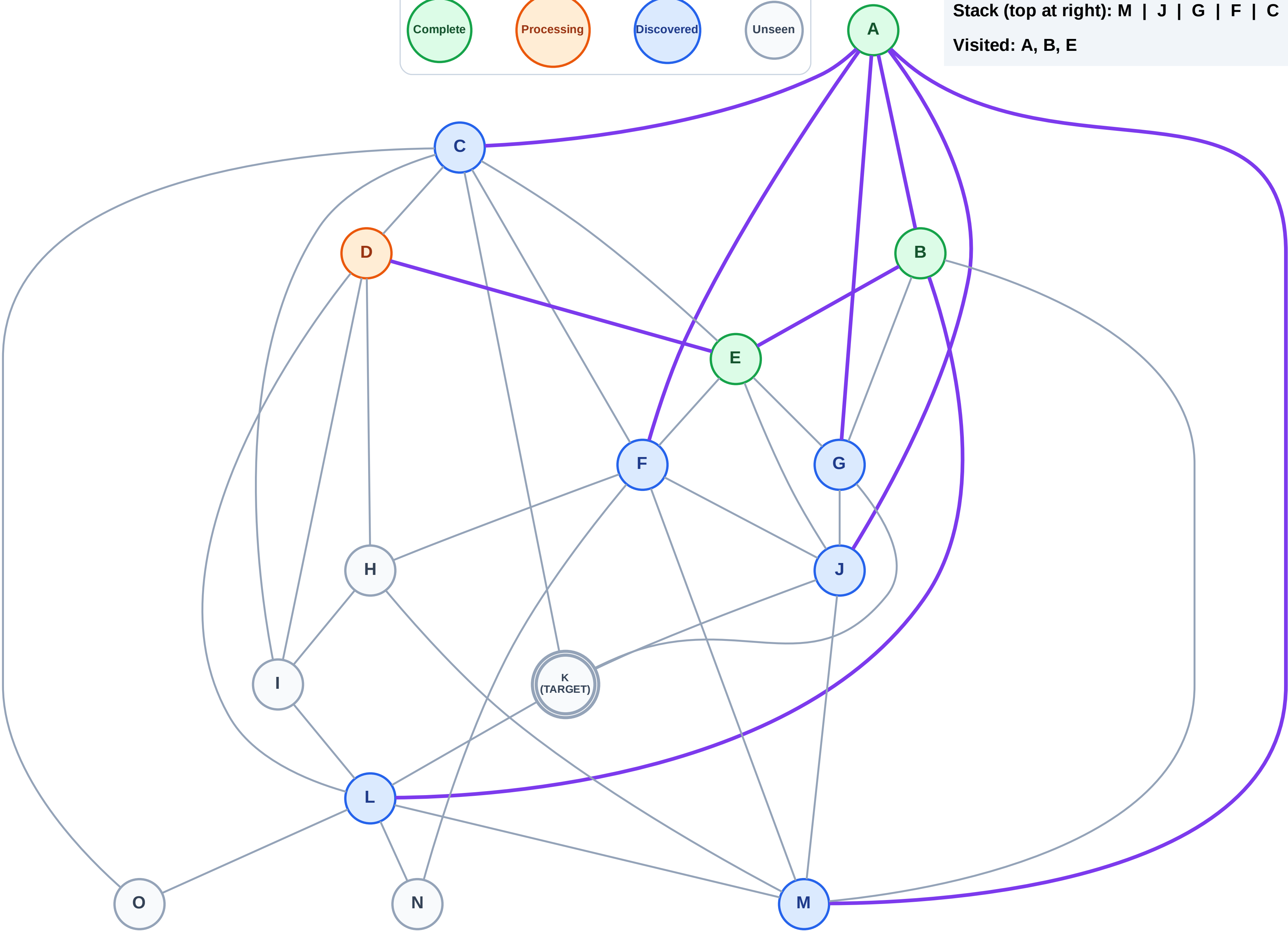
Processing

Discovered

Unseen

Stack (top at right): M | J | G | F | C | L

Visited: A, B, E



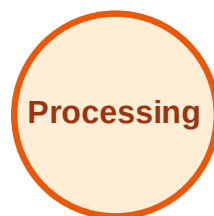
DFS Traversal

Step 9: Discover I, H from D

Legend



Complete



Processing



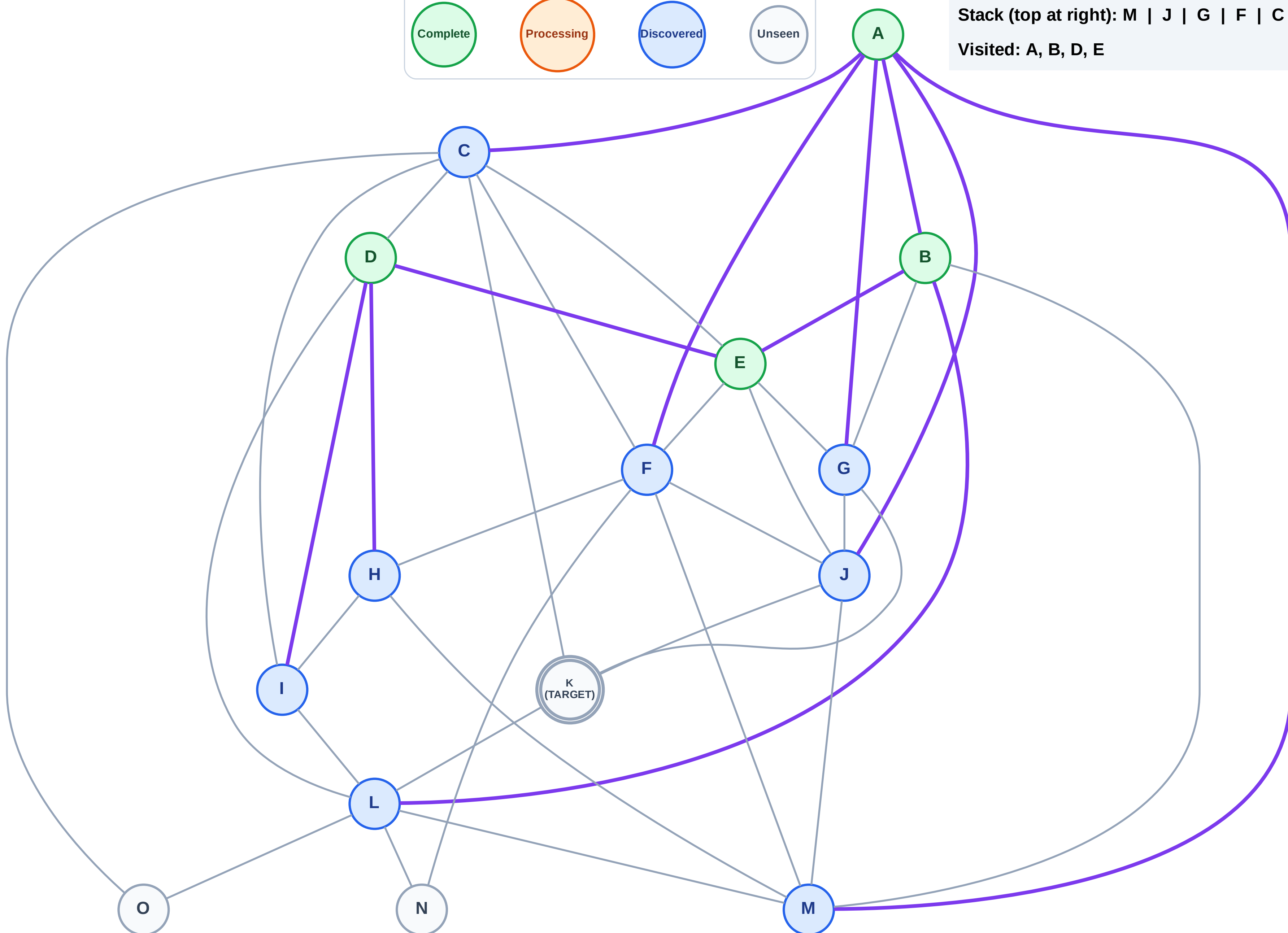
Discovered



Unseen

Stack (top at right): M | J | G | F | C | L | I | H

Visited: A, B, D, E



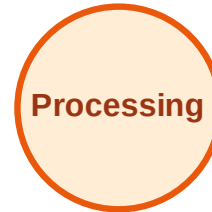
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



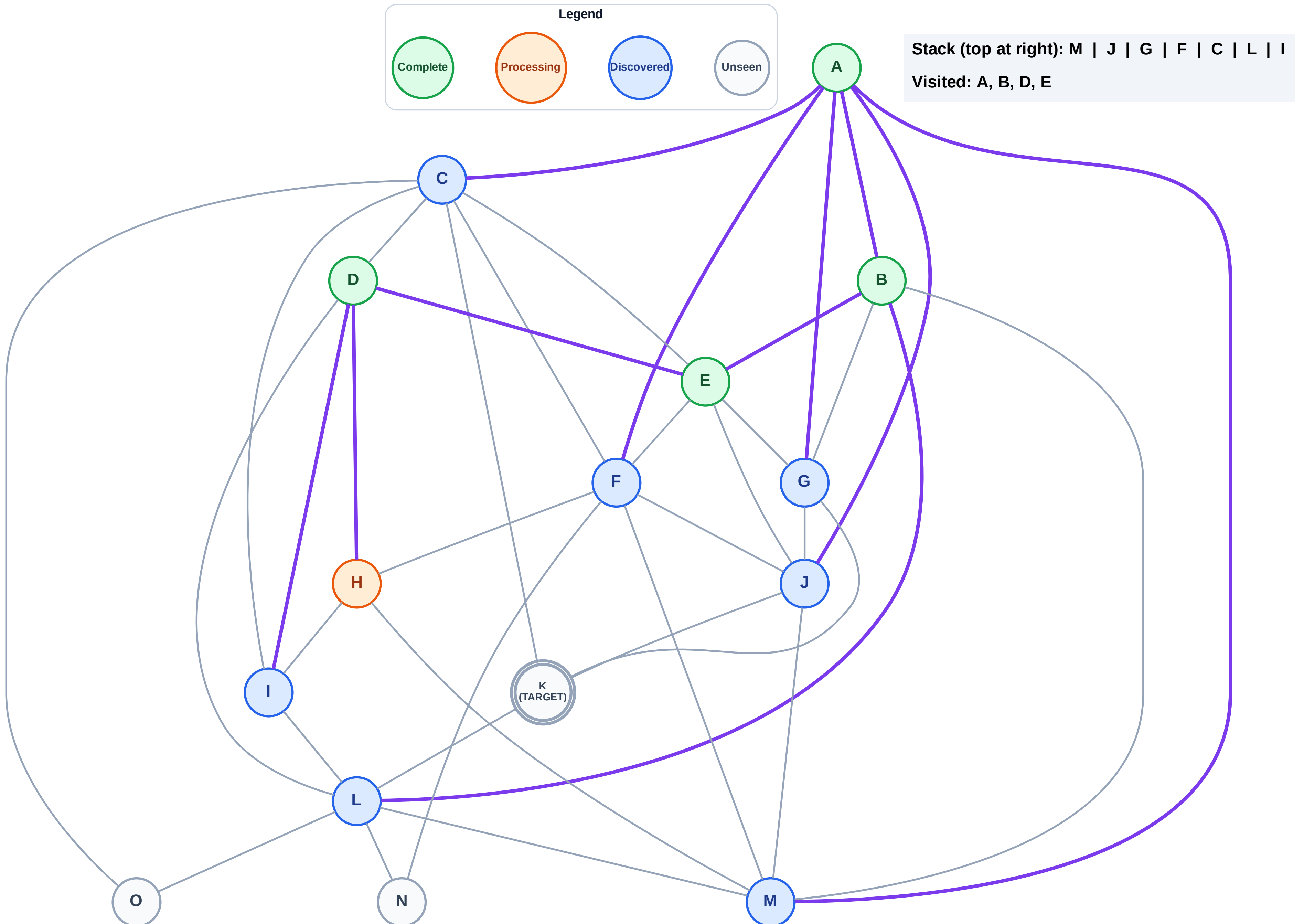
Discovered



Unseen

Stack (top at right): M | J | G | F | C | L | I

Visited: A, B, D, E



DFS Traversal

Step 11: No new neighbors from H

Legend



Complete



Processing



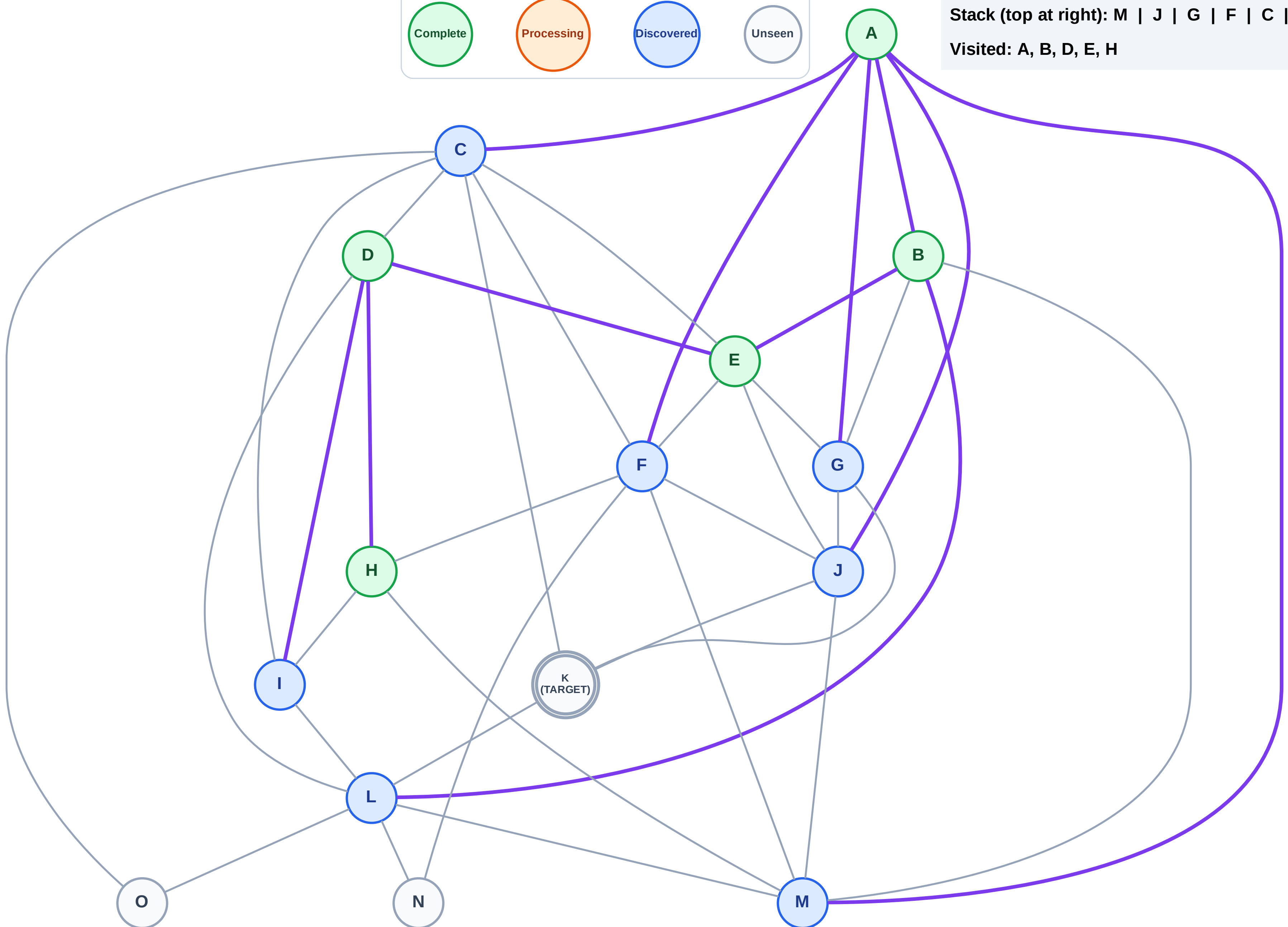
Discovered



Unseen

Stack (top at right): M | J | G | F | C | L | I

Visited: A, B, D, E, H



Step 12: Process node I

Step 12: Process node I

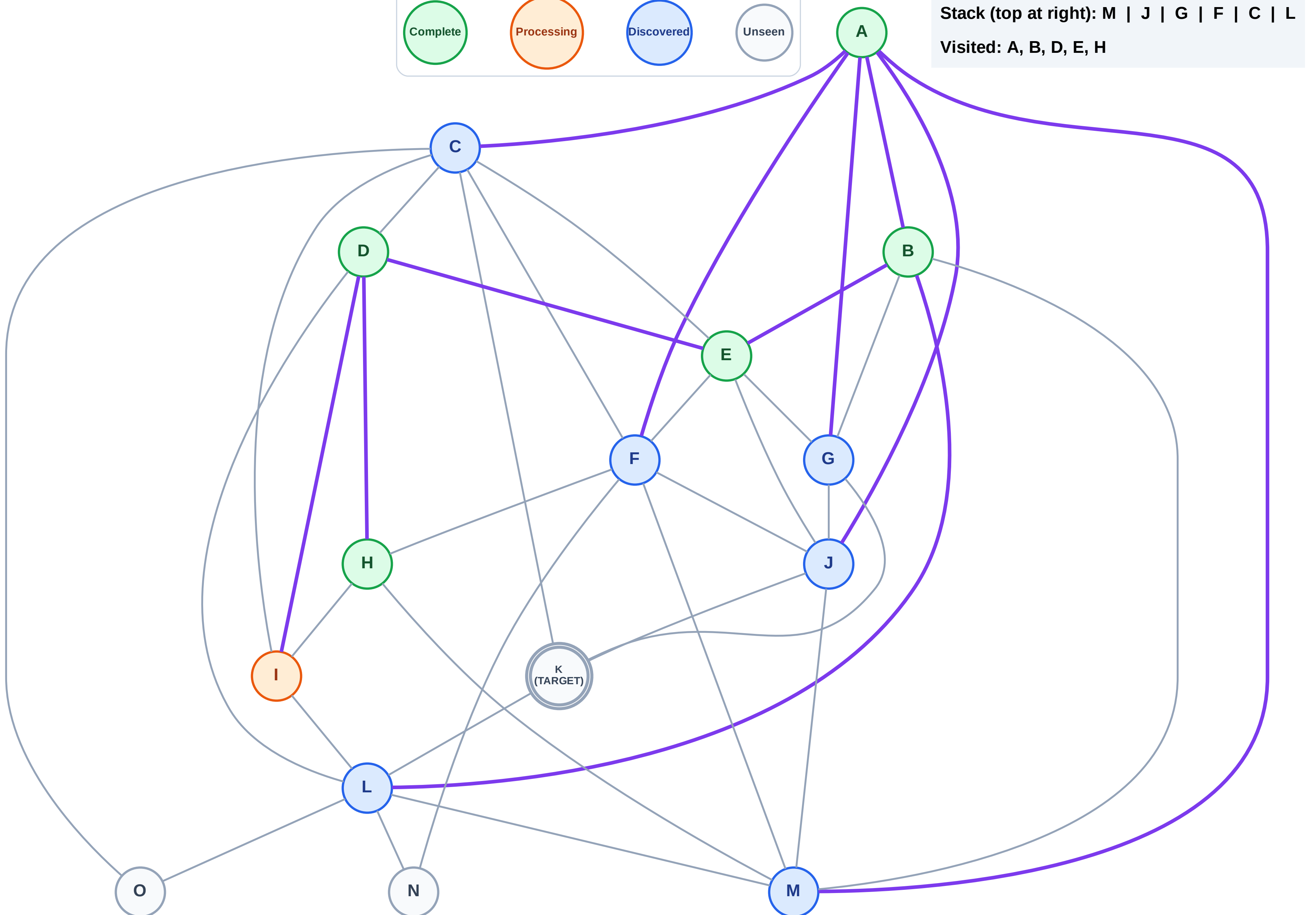
Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

Unseen

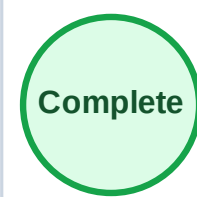
Visited: A, B, D, E, H



DFS Traversal

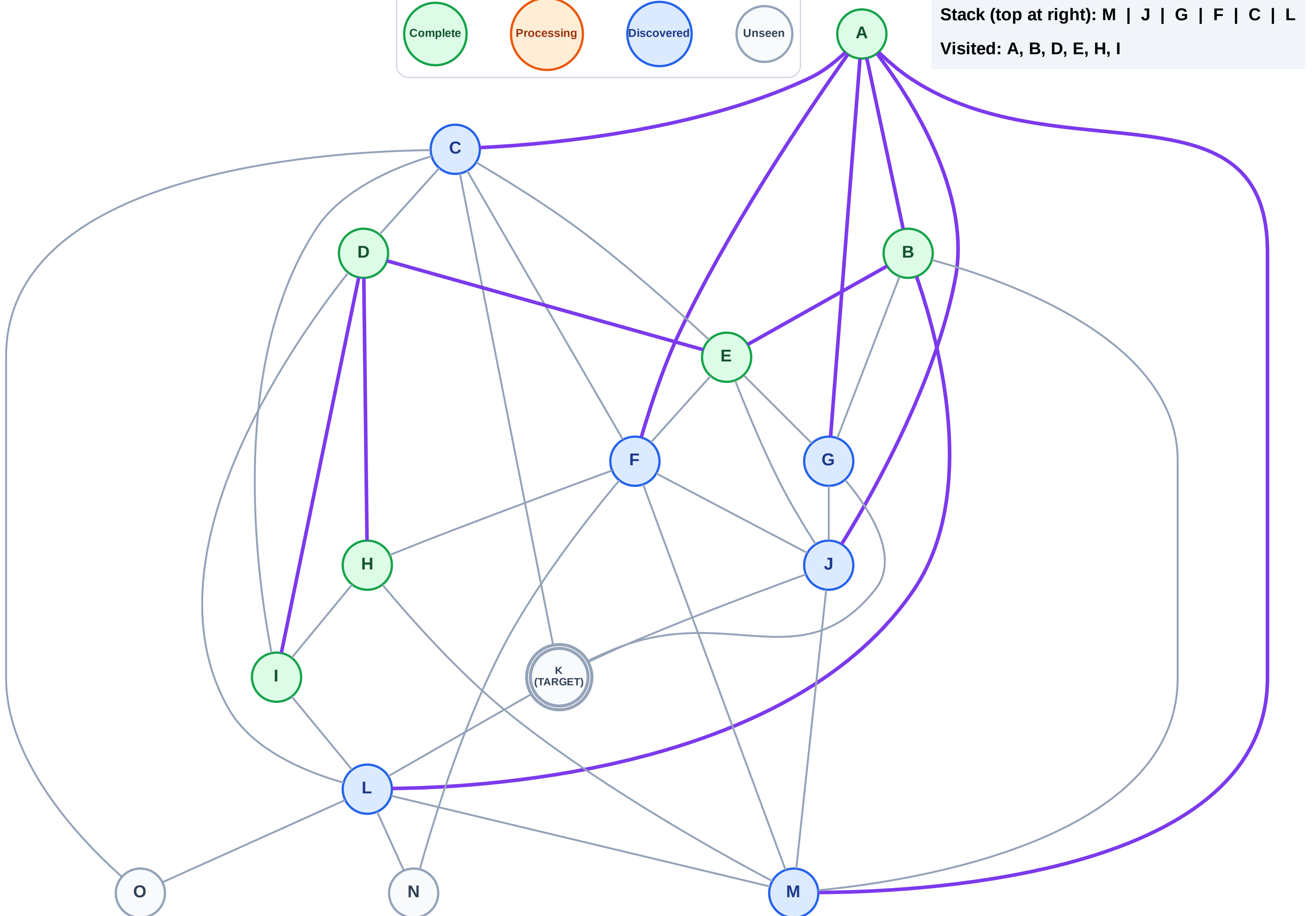
Step 13: No new neighbors from I

Legend



Stack (top at right): M | J | G | F | C | L

Visited: A, B, D, E, H, I



DFS Traversal

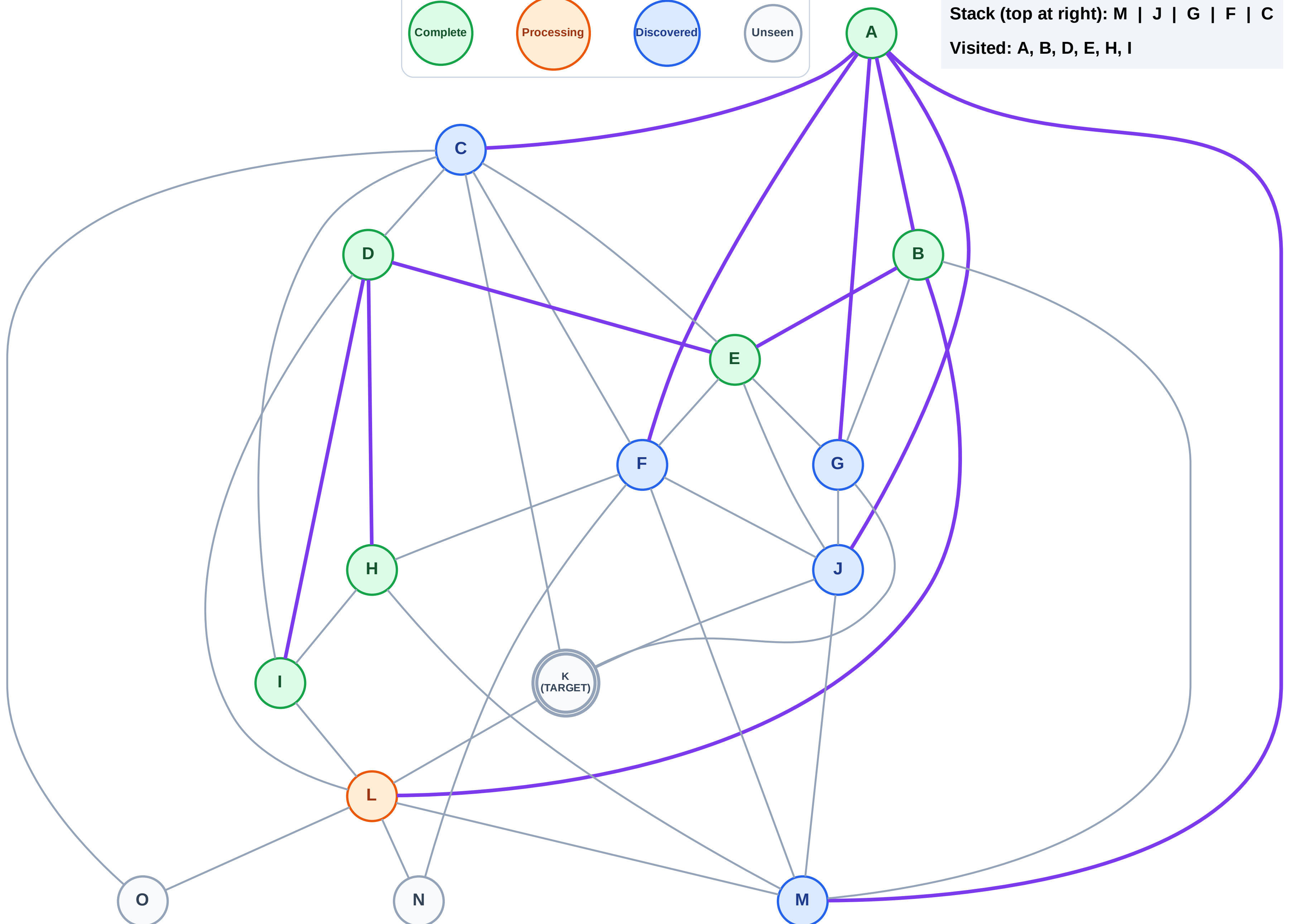
Step 14: Process node L

Legend



Stack (top at right): M | J | G | F | C

Visited: A, B, D, E, H, I



DFS Traversal

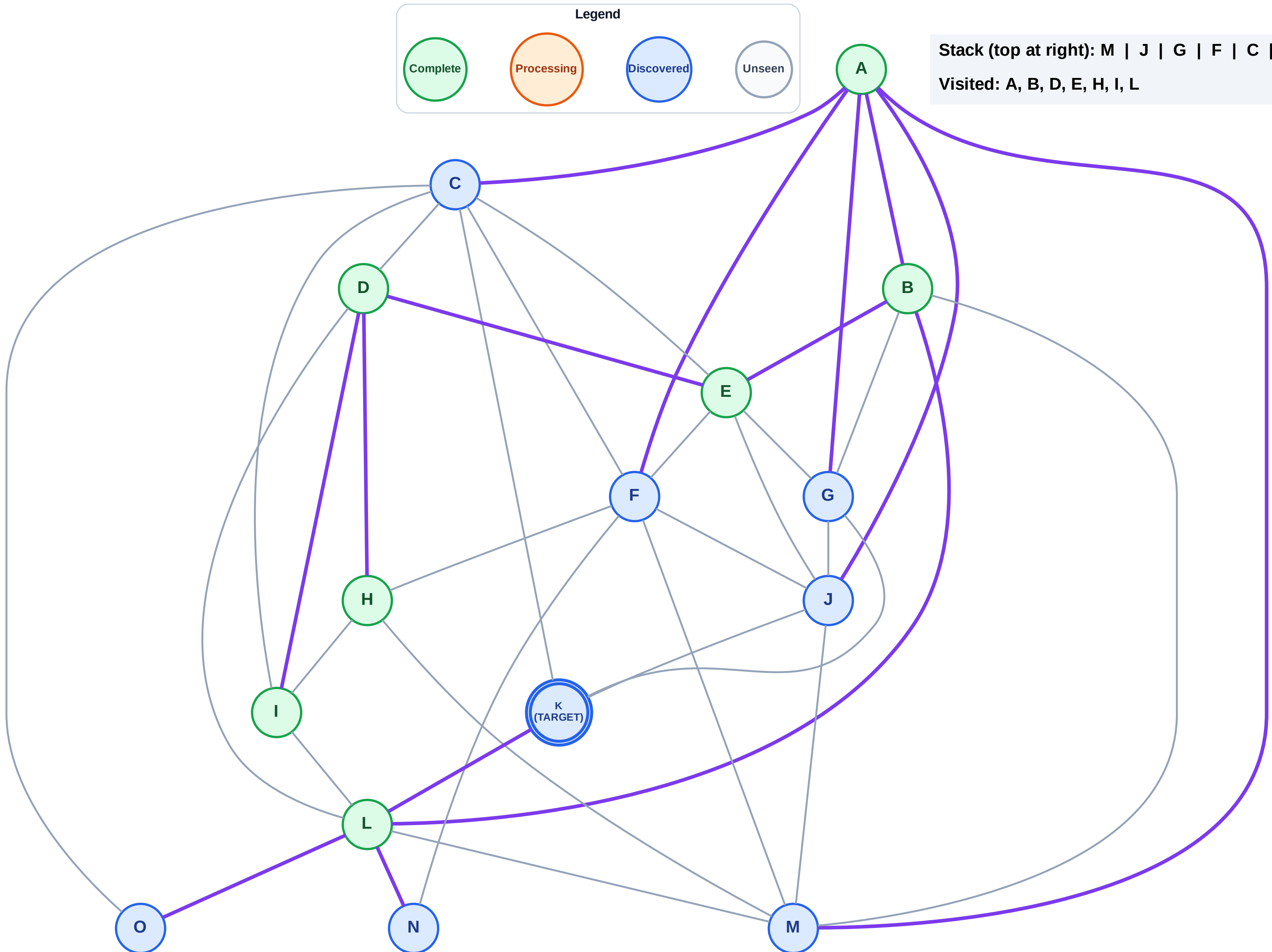
Step 15: Discover O, N, K from L

Legend



Stack (top at right): M | J | G | F | C | O | N | K

Visited: A, B, D, E, H, I, L



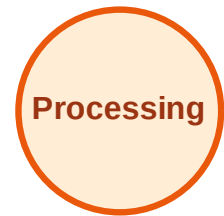
DFS Traversal

Step 16: Process node K

Legend



Complete



Processing



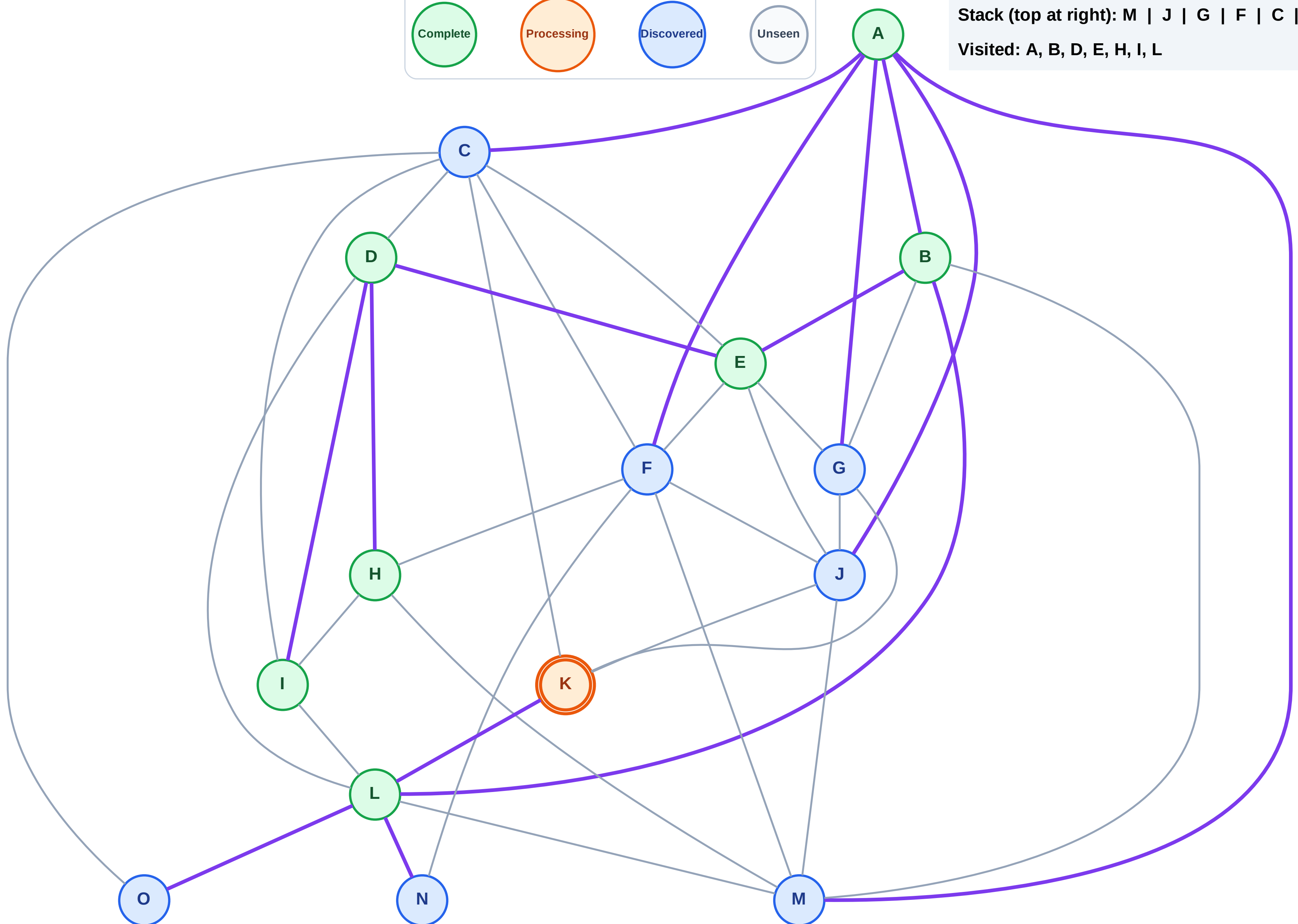
Discovered



Unseen

Stack (top at right): M | J | G | F | C | O | N

Visited: A, B, D, E, H, I, L



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (3 hops): A → B → L → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | J | G | F | C | O | N

Visited: A, B, D, E, H, I, K, L

